

Highlights

Fidelity Is Not Identification: Structure-Aware Calibration of Robot Digital Twins

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- Behavioral fidelity of a robot twin does not certify its physical parameters.
- Dense fits, even with physically consistent inertias, misplace an injected payload.
- Removing the payload exposes the error: 12 mm for dense twins, 0.1 mm for SAPC.
- Selection that admits the uncorrected twin reports no change when none occurred.
- Gazebo and Isaac Sim references and **a physical FR3** confirm the separation.

Fidelity Is Not Identification: Structure-Aware Calibration of Robot Digital Twins

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ABSTRACT

Robot digital twins are validated by how well they reproduce measured trajectories, torques and forces. Inertial parameters differing along structurally null directions behave identically in free space; this paper measures what that ambiguity costs when a twin is edited and resolves it with a sparse-change prior. On a 7-DoF manipulator with exactly known injected discrepancies, dense re-identification, even with physically consistent inertias, and the proposed structure-aware calibration reproduce held-out motions and an unseen press task alike. Yet dense twins attribute an injected 0.8 kg payload to 0.13 kg spread over neighbouring links, whereas the proposed calibration recovers 0.79 kg at the flange; the corrections differ almost only along null directions, which no tested motion separates. Removing each twin's identified payload, the edit available without separate inertia information or re-identification, leaves the dense twins about 12 mm from the unloaded reference, worse than no calibration, and the structure-aware one 0.1 mm. The method selects body and joint corrections from a simulator-derived regressor, admitting the uncorrected twin, and writes them into the executable model. Against Gazebo and Isaac Sim references it recovers the injected changes and Gazebo's missing rotor inertia, and returns no correction where nothing changed. **On a physical FR3 it attributes a weighed 0.81 kg payload to the flange within 3 %, where dense calibration places 0.19 kg, and only its twin remains valid after the removal.** Twin validation should report behavioral fidelity, physical attribution and residual model-form discrepancy separately and test twins under the edits they undergo in service. Code and data: <https://rcilab.khu.ac.kr/sapc>

1. Introduction

Digital twins of robots are simulators that are kept in step with a physical machine so that planning and control algorithms can be tested, tuned and certified before they reach hardware [1, 2]. Everything computed on such a twin rests on one premise: that its dynamics have been calibrated to the robot it stands for. In practice calibration is done by matching behavior. Simulator parameters are tuned until the twin reproduces measured trajectories, torques or disturbance responses [3, 4, 5], or the rigid-body parameters are re-identified from excitation data under physical-consistency constraints [6, 7]. Credibility frameworks tie the evidence a model needs to its context of use and to the risk of the decision it informs [8], and for robot simulators that evidence is overwhelmingly behavioral: validation frameworks [9, 10] and credibility scores for twins [11] compare the outputs of the model with measured outputs, and comparisons of simulators rank them by the accuracy of the motion they produce [12, 13]. A twin is therefore usually declared faithful when it behaves like the robot.

Behavior, however, does not determine the physical parameters. Fig. 1 shows how the question arises: the torque that moves the MuJoCo TCP 10 cm downward produces a different motion in each of three simulators, and a calibration that restores the motion must decide where the correction goes. The rigid-body dynamics of a manipulator depend on its inertial parameters only through a reduced set of base combinations, and the remaining directions of parameter space produce no torque in any motion [14, 15, 16]. Two twins whose parameters differ along those directions behave identically in free space, on the calibration data and on every other trajectory, so a calibration that is judged by behavior alone is free to place a physical change in the wrong body. That a model can predict well while its physical parameters are wrong is familiar from the calibration of computer models with discrepancy [17, 18]; for a robot twin it has a concrete consequence. The questions such a twin is kept for, which component changed, by how much, and what happens when a component is exchanged, are questions about its parameters. A twin whose payload has been spread

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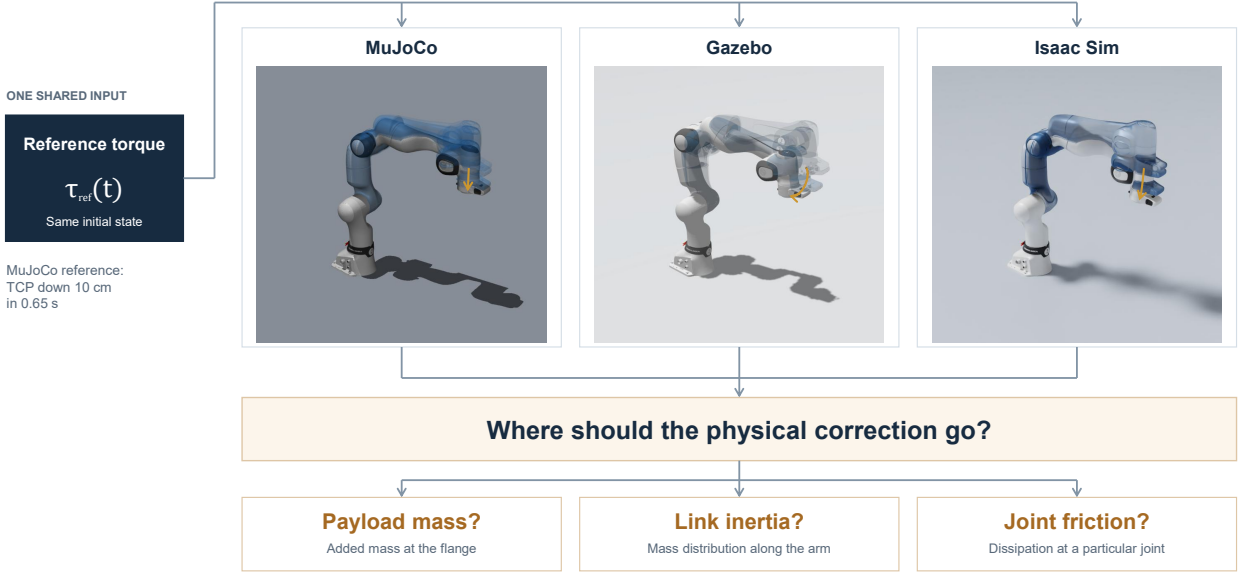


Figure 1: Where should the physical correction go? The reference torque $\tau_{ref}(t)$ is the torque that moves the TCP of the MuJoCo FR3 10 cm downward in 0.65 s. Applied from the same initial state to MuJoCo, to Gazebo and to Isaac Sim, the last with a 100 g payload added at the flange, it produces a different motion in each simulator: at 0.65 s the Gazebo TCP is 82 mm and the Isaac Sim TCP 46 mm from the MuJoCo TCP. Ghosts show the robot at 0.000, 0.217 and 0.433 s, the solid pose is at 0.650 s, and the orange line is the measured TCP path. A calibration that restores the motion must decide whether the correction belongs to the payload, to the mass distribution along the arm, to the friction of a particular joint or to a property that one simulator does not represent, such as the rotor inertia absent from the Gazebo robot description, and reproducing the motion does not by itself answer this question.

over neighbouring links reproduces every measured motion, and the error surfaces when the twin is edited, most clearly when the tool it identified is taken off.

This paper makes this failure measurable and introduces a calibration protocol that scores physical attribution alongside behavioral fidelity. We build a controlled study around a 7-DoF manipulator whose reference dynamics carry known, deliberately injected physical discrepancies, a payload on the flange, friction changes on two joints and a mass change on one link, so that every calibration can be scored both on how well its twin behaves and on whether it found the injected change. On this test bed a dense re-identification of all parameters, also when it is constrained to physically consistent inertias, and the calibration proposed here reproduce held-out motion and an unseen contact task alike, yet only the proposed calibration attributes the discrepancy to the payload that was actually added, and only its twin remains correct once the payload it identified is removed. **A physical FR3 repeats the result with a weighed payload: from the robot's own joint-torque sensors the proposed calibration attributes it to the flange, and dense calibration does not.** The proposed method, Structure-Aware Physical Calibration (SAPC), treats calibration as physical attribution under an explicit prior: relative to an existing simulator, a change is concentrated in few bodies and joints. It extracts its regressor from the simulator itself by perturbing physically valid bodies, so that the identification model is the model the twin executes. It then selects corrections at the level of bodies and joints, with the uncorrected twin as one of the candidates and under bounds that keep the corrections dissipative and the masses positive, and it writes the selected correction back into the native parameter fields of the simulator, so that the result is an executable calibrated twin and not a residual function. Because some bodies cannot be separated from their neighbours by the excitation, the method also reports how stable each attribution is under repeated measurement noise and which alternatives the data cannot distinguish, instead of forcing one answer.

The same pipeline is evaluated with references produced by two other simulators, Gazebo and Isaac Sim, while the twin stays in MuJoCo. Cross-simulator calibration adds differences in how simulators represent a robot and integrate its dynamics. Some of these differences are representable by parameters of the twin and are then recovered as such, the absence of rotor inertia in the SDF description used by Gazebo being the clearest example; the rest remains after calibration and is reported on its own, measured against a twin that carries the true parameters of the reference.

The contributions are four. First, we turn the known structural non-identifiability of inertial parameters into a measurable validation question: on references with known physical discrepancies, accurate held-out motion and unseen contact prediction coexist with incorrect component-level attribution, also for physically consistent dense identification, and we delimit when the attribution error matters for editing the twin. Second, we develop a simulator-native calibration pipeline that couples bounded body and joint selection, physical-consistency projection and executable parameter write-back. Third, we quantify attribution stability under measurement-noise resampling, relate ambiguous link assignments to the structural identifiability of the arm, and test the selection on discrepancy patterns other than the main one, including a distributed pattern that violates its prior. Fourth, we evaluate the same pipeline against MuJoCo, Gazebo and Isaac Sim references, separating injected corrections, representable differences between simulator descriptions and the residual that remains relative to a twin with the true parameters, **and we confirm the payload attribution on a physical FR3**.

Section 2 positions the work, Section 3 defines the three quantities in which a twin is judged, Section 4 presents SAPC, Section 5 reports the experiments, and Sections 6 and 7 discuss and conclude.

2. Related work

Four bodies of work bear on calibrating a robot digital twin: classical dynamic identification, sparse and physics-informed identification, real-to-sim calibration and hybrid twins, and the validation of simulation models. Their evaluations center on reproducing behavior or on recovering a physically valid parameter vector. Whether the correction lands on the body or joint that actually changed is rarely scored separately, and it is the question this paper isolates.

2.1. Classical robot dynamic identification

Classical identification writes rigid-body inverse dynamics as a regressor that is linear in the standard inertial parameters and fits them to data from optimised excitation trajectories [15]. Only base parameters, the combinations that affect joint torques, can be identified from motion [14], and which inertial quantities are observable, including the transfer of inertia between neighbouring links, has been characterised geometrically [16]. Physical consistency is enforced by linear matrix inequalities on the pseudo-inertia matrix [19, 7] or by penalty-based retrieval of feasible parameters, as done for the Franka Panda [6], and a payload is estimated from motion when it is known to sit at the end effector [20]. These methods return a physically valid parameter vector that reproduces the torques. When several components may have changed, however, a physically valid solution is not unique, and the correction can be distributed over bodies whose torque signatures alias with the one that changed. SAPC keeps the linear regressor and the consistency conditions and addresses a complementary calibration question: which body or joint changed relative to an existing simulator, and how stable that attribution is when several corrections achieve the same predictive accuracy.

2.2. Sparse and physics-informed identification

Sparse regression selects a few active terms from a library of candidate functions [21], and physics-informed variants train a neural surrogate of the state jointly with the sparse model so that derivatives need not be estimated from noisy samples [22]. In robotics these tools learn the discrepancy between a nominal model and the plant as friction and structural-uncertainty terms for fault-tolerant control [23], recover interpretable structural terms of manipulator dynamics [24], and preserve mechanical structure while adapting inertial, Coriolis and generalized-force corrections under Bayesian sparsity [25]; physics-informed surrogates trained on simulator data have also been proposed for contact-rich robot dynamics [26]. Sparsity in these works is imposed on individual library terms, and over an aliased inertial library an elementwise support is not unique. SAPC imposes sparsity on physical groups, selects them by cross-validated forward selection under parameter bounds and scores the resulting attribution against known injected changes; a derivative surrogate adapted from [22] is compared with filtered derivatives.

2.3. Real-to-sim calibration and hybrid twins

Real-to-sim calibration tunes simulator parameters until simulated observations match measured ones. TuneNet learns a one-shot residual update of the parameters from a single real observation [4], friction, inertial and gravity terms of a Panda have been tuned in Gazebo and MuJoCo by genetic search [5], and differentiable simulators have been augmented with neural residuals that absorb what the parameters cannot [27]. Hybrid digital twins couple a physics simulator with data-driven submodels and add a confidence measure that decides when the learned part is trusted [28]. These approaches are judged by transfer performance, and a neural residual cannot be written back into

the physical parameters of the simulator. SAPC writes its correction into the native fields of the simulator wherever the simulator represents the term, and it reports what remains separately.

2.4. Validation of simulation models and digital twins

The credibility of a simulation model has long been established through conceptual, operational and data validity [29], and current standards scale the required evidence with the context of use and the decision risk [8]. For robot simulations, validation has been formalised as a framework that compares the simulator with the real robot as a whole [9], modular frameworks combine analytical and simulation checks for manufacturing systems [10], and evidence theory aggregates the credibility of a twin into a score [11]. Robot digital-twin work in this journal compares simulators [12, 13] and validates multibody co-simulation twins by their output trajectories [3]; twin architectures quantify uncertainty on their outputs [30], serve as test platforms whose conclusions inherit the twin's calibration [2], and are recalibrated over time to follow a drifting physical system [31]. In the calibration of computer models, Kennedy and O'Hagan introduced an explicit model-discrepancy term [17], and Brynjarsdóttir and O'Hagan showed that without an adequate discrepancy model a calibration can predict well while returning biased physical parameters [18]. This paper turns that separation into an operational protocol for robot digital twins: references with exact physical ground truth, within one simulator and across simulators, in which behavioral fidelity, physical attribution and residual model-form discrepancy are measured separately, and in which the calibrated twin is executed, edited and transferred.

3. Problem formulation and fidelity decomposition

A calibrated twin is judged here in three separate coordinates: whether it reproduces what the reference does, whether its correction sits in the right bodies and joints with the right magnitude, and what remains when every injected parameter is known. This section fixes the setting and defines the three quantities.

3.1. Nominal twin, reference system and physical discrepancy

Let T_0 be the nominal simulator of the robot, with inverse dynamics $ID_0(q, \dot{q}, \ddot{q})$, and let R be the reference system that generates the observations. In the controlled study of Section 5, R is a simulator whose physical ground truth is exact: either the same simulator with injected discrepancies (within-simulator controlled validation) or a different simulator built from the same model description (cross-simulator validation), Gazebo or Isaac Sim; in every case the twin is MuJoCo [32, 33]. The reference tracks an excitation trajectory under a nominal-feedforward-plus-PD controller and logs joint positions q and torques τ at 1 kHz, from which $\dot{q}$ and $\ddot{q}$ are estimated by Savitzky-Golay differentiation. The physical discrepancy is the residual

$$r(q, \dot{q}, \ddot{q}) = \tau - ID_0(q, \dot{q}, \ddot{q}), \quad (1)$$

which calibration expresses through a library of candidate physical corrections, $r \approx A(q, \dot{q}, \ddot{q})\xi$. The inertial columns of A are extracted from the simulator itself (Section 4) and the friction and rotor-inertia columns are analytic functions of $\dot{q}$ and $\ddot{q}$, so A is a simulator-native regressor; its columns are partitioned into groups $\mathcal{G}$ by the body or joint they modify, ten standard inertial parameters [15] for each of the seven links, five friction terms for each joint and one rotor-inertia (armature) term per joint. All corrections are increments relative to T_0 .

Calibration in this setting updates an existing simulator, and we assume that the discrepancy relative to T_0 is concentrated in few groups: the situation of a twin that tracks a payload change, the wear of a joint or the replacement of a component. This sparse-change assumption is the prior that makes attribution possible (Section 4.6); Section 5.5 tests it with discrepancy patterns other than the main one, including a distributed pattern that violates it. The output of any calibration is a triple (S, ξ_S, T_c) : the selected groups $S \subseteq \mathcal{G}$, the magnitudes ξ_S of their corrections, and the executable calibrated twin T_c obtained by writing ξ_S back into the native fields of the simulator. Dense least squares returns $S = \mathcal{G}$, a structure-aware calibration returns a small S , and both return a T_c that runs; the question is which component of the triple a given measurement validates.

3.2. Behavioral fidelity

Behavioral fidelity is what digital-twin validation reports almost exclusively: an output of the twin against an output of the system [9, 3], or a task-level outcome [13].

Definition 1 (Behavioral fidelity). Let r be the residual (1) on held-out trajectories that were not used for calibration. *Torque fidelity* (F1) is reported twice. The linear-library NRMSE, $\text{RMS}(r - A\xi)/\text{RMS}(r)$, measures how well the fitted coefficients explain the residual. The executable-twin NRMSE, $\text{RMS}(\tau - \text{ID}_{T_c})/\text{RMS}(r)$, evaluates the inverse dynamics of the instantiated twin T_c , after projection and native write-back, on the same samples. Both equal 1 for the nominal twin, and their denominators are specific to each reference, so values are compared within a condition. *Trajectory fidelity* (F2) is the tool-centre-point (TCP) RMSE when T_c runs the same controller on the same reference trajectory as R (closed-loop replay); as a secondary measure, the open-loop divergence is the TCP error at the end of 0.2 s windows in which the torque logged on R is replayed from the state of R . *Contact fidelity* (F3) is the contact-onset time, the holding force during the hold phase and the impulse of the contact force in a press task that is not used for calibration, together with the TCP RMSE during the press.

Two anchors fix the scale of these metrics: the nominal twin, which carries none of the discrepancy, and a twin that carries the true parameters of the reference, which gives the scale of what the twin cannot represent. Peak force and pointwise force-waveform RMSE are excluded from F3 because, as Appendix D shows, they are properties of the integrator and of contact chatter as much as of the model. Every quantity in Definition 1 compares outputs; none of them examines ξ .

3.3. Physical attribution fidelity

Physical attribution asks whether the correction sits in the bodies and joints that actually changed, with the right magnitude, and how stable that answer is.

Definition 2 (Physical attribution). Let $S^* \subseteq \mathcal{G}$ be the groups whose parameters differ between R and T_0 and ξ^* their true increments. *Support attribution* is the precision and recall of the selected set S with respect to S^* . *Magnitude attribution* is the error of the physical quantities reconstructed from ξ_S : payload mass and centre of mass, link mass, and the Coulomb and viscous increments of each joint. *Selection stability* is the frequency with which each group enters S when the calibration is repeated on the same trajectories with independently resampled measurement noise. The *empirical group-separation diagnostic* is the rank of the restricted regressor A_G of a group G on the excitation data, together with the fraction of the standardised torque signature of one group that lies in the column space of another.

These quantities can be scored only when S^* and ξ^* are known, which is why the reference is a controlled simulator with injected discrepancies. Selection stability is a noise-resampling measure on a fixed excitation, and a frequency of 10 in 10 is not a probability of being correct. SAPC reports attribution at the resolution the data support: stable body and joint selections, together with the directions the excitation cannot resolve. Behavioral fidelity constrains a correction only up to the null space of the regressor, and Section 4.6 shows that for rigid-body parameters this null space is structural, so F1 and F2 are blind to the difference between two corrections in every free-space motion, not only on the visited states. Attribution is nevertheless what a twin is maintained with, which component drifted and by how much [31], and it is what makes a credibility score [11] physically interpretable.

3.4. Residual model-form discrepancy

Not every discrepancy between R and T_0 has a column in A . The residual model-form discrepancy is measured by the behavioral fidelity of the twin $T_c(\xi^*)$ that carries the true parameters of R ; this matched-parameter twin gives the scale of what calibration cannot remove, and it is not an additive decomposition of the errors of other twins. In the controlled design it is measured directly, and it has four sources.

Numerical integration. The twin and the reference may integrate with different schemes and steps; in free space the effect is small.

Friction regularisation. Simulators implement Coulomb friction as a regularised constraint whose force is linear in velocity below a knee and saturates above it, while the library uses a smooth analytic Coulomb term. The two agree on the saturated branch and differ near the knee, so identification rows near zero velocity are excluded, and the knee contributes to the linear-library residual of every scheme; an executable twin that uses the simulator's own friction law does not inherit this contribution.

Contact and constraint solver. Contact laws and their solvers differ between simulators, so two simulators with the same robot parameters do not produce the same contact forces; this part of the discrepancy is not of the robot and cannot be corrected by robot parameters.

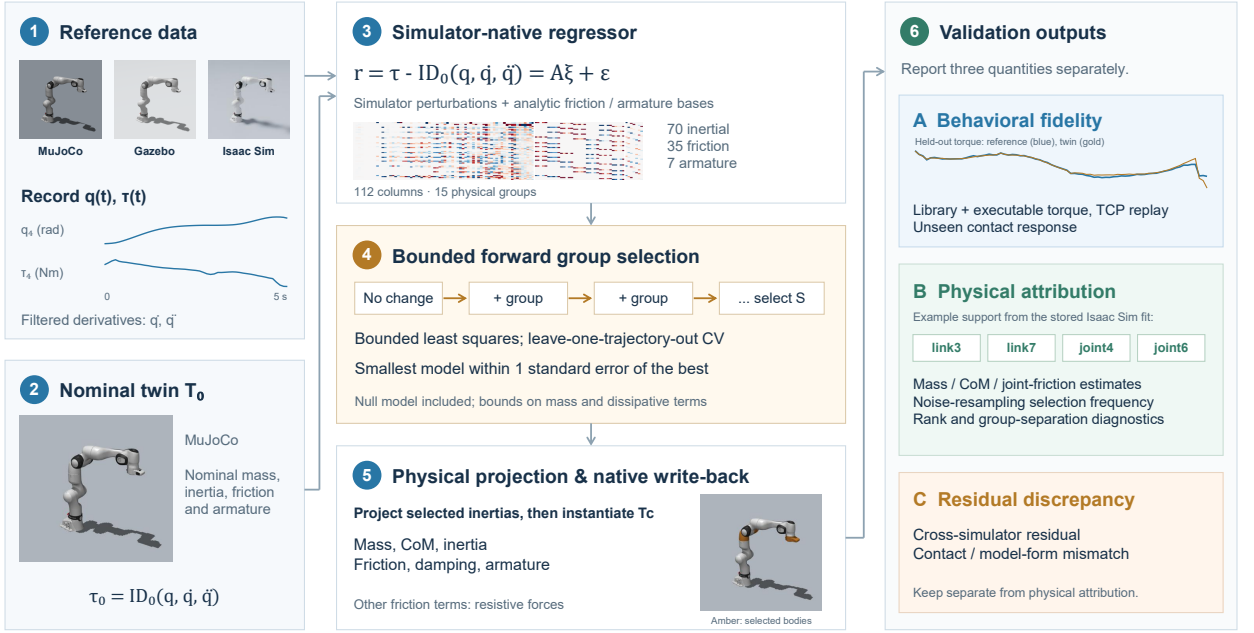


Figure 2: Overview of structure-aware physical calibration (SAPC). (1) Joint positions and torques are recorded on a reference simulator, MuJoCo with injected discrepancies, Gazebo or Isaac Sim, and velocities and accelerations are obtained by filtered differentiation. (2) The nominal MuJoCo twin T_0 , with nominal mass, inertia, friction and armature, gives $\tau_0 = ID_0(q, \dot{q}, \ddot{q})$. (3) The residual $r = \tau - ID_0$ is expressed in a simulator-native regressor with 70 inertial, 35 friction and 7 armature columns in 15 physical groups. (4) Bounded forward selection over these groups, starting from the uncorrected twin, chooses the smallest model within one standard error of the best leave-one-trajectory-out score. (5) The selected inertias are projected to physically consistent values and written, with friction, damping and armature, into the native fields of the executable twin T_c ; the remaining friction terms act as resistive forces. (6) Behavioral fidelity, physical attribution and residual discrepancy are reported separately.

Simulator-specific representation. A term present in one simulator and absent from the other is a representable parametric discrepancy whenever A has a column for it, and it is recovered as such; what remains is specific to the simulator pair, the library and the excitation rather than a universal floor.

Reporting the residual separately prevents two errors that a single fidelity number invites: a regression fits part of the model-form discrepancy with parameters and spreads it over nuisance groups, turning model-form error into false attribution, and a small behavioral residual is read as evidence that the parameters are right.

4. Structure-aware simulator-native calibration

SAPC converts the residual $r = \tau - ID_0$ of Section 3 into a correction of the simulator's own physical parameters. It extracts the inertial regressor from the simulator (Section 4.1), adds a friction and rotor-inertia basis (Section 4.2), bounds every candidate fit (Section 4.3), selects body and joint groups by cross-validated forward selection (Section 4.4) and writes the result back into native simulator fields (Section 4.5); Fig. 2 summarises the pipeline. Section 4.6 states why the behavioral fidelity of the outcome is not evidence of its physical attribution.

Three facts of rigid-body identification recur below. Joint torques depend on the 70 standard inertial parameters of the arm only through 43 combinations, the base parameters, because some parameters never produce a torque, such as a mass placed on a joint axis that does not move with that joint, and others enter only in fixed sums with those of a neighbouring link; a change along the remaining 27 directions is invisible in every motion [14, 15]. A set of inertial parameters describes a realisable rigid body only if its pseudo-inertia matrix is positive semidefinite, a condition that a linear matrix inequality (LMI) imposes on a fit [7]. The armature is the rotor inertia of a joint's motor reflected through its gearbox, which adds a term proportional to that joint's acceleration to that joint's torque alone.

4.1. Simulator-derived inertial regressor

For a rigid body b with standard parameters $\pi_b = (m, m\mathbf{c}, \mathbf{I}_o)$, mass, first moment and inertia about the body origin, inverse dynamics is affine, $\text{ID}(\pi_b + \delta) = \text{ID}(\pi_b) + Y_b(q, \dot{q}, \ddot{q})\delta$, and rotor inertia and joint damping do not depend on π_b [15]. Classical identification derives Y_b symbolically from the kinematic chain [15, 6]; we read it off the simulator instead, so the regressor is by construction the model the twin will execute.

Lemma 1 (π -basis extraction). *Let $P = [\pi_1 \ \dots \ \pi_{10}]$ be the standard parameter vectors of ten physically valid rigid bodies with P invertible. Then for any body b and any state $(q, \dot{q}, \ddot{q})$,*

$$Y_b(q, \dot{q}, \ddot{q}) = [\text{ID}(\pi_b + \pi_1) - \text{ID}(\pi_b), \dots, \text{ID}(\pi_b + \pi_{10}) - \text{ID}(\pi_b)] P^{-1}, \quad (2)$$

where every term on the right is an inverse-dynamics call of the simulator with a physically valid body.

The sum of two valid bodies is a valid body, so the simulator accepts every $\pi_b + \pi_k$, and substituting the affine form into the bracket gives $Y_b P P^{-1} = Y_b$. A link's regressor thus costs eleven inverse-dynamics calls and requires only that the simulator expose inverse dynamics and writable inertial fields. Our basis bodies have mass 1 kg, centre-of-mass offsets of 0.1 m and principal inertias of 0.01-0.02 kg m² with and without a 45° rotation, which gives P a condition number of 942. A payload rigidly attached to the flange is a change of the ten parameters of link 7, so the library needs no separate payload body.

The extracted regressor reproduces the torque change of the injected payload and link-3 mass to a maximum error of 1.4×10^{-5} N m over a 9.6 N m range. Extraction is performed with the simulator's friction-loss constraint disabled; with it enabled the error grows to 0.085 N m, because that constraint force depends on the effective inertia, a velocity effect that Section 4.2 handles. Of the 70 inertial parameters of the seven links, only 43 combinations affect the joint torques in any motion: 11 columns vanish identically and the other 59 span the 43-dimensional base-parameter space of the arm [14, 15]. This is why an elementwise support is not unique, and why attribution is made per body and joint (Section 4.4).

4.2. Friction and armature basis

Each joint j contributes five columns acting on its own equation: Coulomb $\tanh(\dot{q}_j/0.02)$, viscous $\dot{q}_j$, Stribeck $\tanh(\dot{q}_j/0.02) \exp(-(\dot{q}_j/0.1)^2)$, quadratic $\dot{q}_j |\dot{q}_j|$ and a constant offset, with $\dot{q}_j$ in rad/s. One further column per joint, $\ddot{q}_j$ on row j , represents a change of rotor inertia (armature); the seven armature columns form one simulator-level group, because a reference description without rotor inertia, such as SDF, changes all of them at once (Appendix C). The library thus has 112 columns in 15 groups.

The simulator's own Coulomb friction is a constraint, not a function: its force is linear in velocity below a knee $v^* = f/(M_{\text{eff}}b)$, about 0.03 rad/s at joint 4, and saturates at $\pm f$ beyond it. The tanh column matches the saturated regime to 5×10^{-5} N m but misses by up to 1.8 N m at the knee, whose width depends on the mass matrix and is deliberately not in the library. As in identification practice [6], the equation of joint j is dropped from the regression wherever $|\dot{q}_j| < 0.1$ rad/s, a cut-off wide enough that the knee does not leak into the viscous coefficient. The knee is a residual model-form discrepancy of the linear library in the sense of Section 3: it contributes to the library residual of every scheme, whereas the executable twin, which writes Coulomb increments into the simulator's own friction-loss field, reproduces it.

4.3. Bounds and physical consistency

Candidate fits satisfy mass and dissipative-coefficient bounds. The total Coulomb, viscous, Stribeck and quadratic coefficients of each joint, nominal plus increment, and the total rotor inertia of each joint must be non-negative, and no body may lose more than half its nominal mass; the constant offset is free. These are box constraints on the increments, enforced inside every fit by bounded-variable least squares on standardised columns. They are not optional: without them a sparse fit on a short record can return non-passive friction coefficients, and the twin built from them diverges.

Full physical consistency of the inertial parameters is imposed after selection. A set of inertial parameters belongs to a realisable rigid body if and only if its pseudo-inertia

$$J(\pi) = \begin{bmatrix} \frac{1}{2} \text{tr}(\mathbf{I}_o) \mathbf{1} - \mathbf{I}_o & \mathbf{h} \\ \mathbf{h}^\top & m \end{bmatrix}, \quad \mathbf{h} = m\mathbf{c}, \quad (3)$$

is positive semidefinite [7]. A selected body that violates this condition is replaced by the nearest consistent body in the Frobenius norm of J , obtained by clipping the eigenvalues of J at 10^{-7} with a mass floor of 1 g, and the relative projection distance is reported. This projection implements the consistency condition; it is not the estimator of [19, 7], which imposes the matrix inequalities inside the fit, and Section 5.2 compares SAPC with such an estimator. Projection follows group selection rather than a dense fit, because projecting an aliased dense solution alters the torque function it was fitted to.

4.4. Forward group selection

Attribution is made over 15 groups: the ten inertial columns of each of the seven links, the five friction columns of each of the seven joints and the seven armature columns. A group is the unit at which a physical change occurs in practice, a payload, a worn joint, a replaced link, and the unit at which a twin drifts from its physical counterpart in service [31].

SAPC selects groups greedily, starting from the uncorrected twin, whose cross-validated score is the RMS of the residual itself. Each unselected group is tentatively added, its columns are fitted by bounded least squares with a ridge penalty of 10^{-4} in standardised units, and the leave-one-trajectory-out cross-validated torque RMSE is computed over the training trajectories. The group that lowers the score most is added, and the search stops when no addition lowers it, or at eight groups. Among the models on the path, the uncorrected twin included, the smallest whose score lies within one standard error of the best is chosen (1-SE rule; the standard error is the sample standard deviation of the fold scores divided by the square root of the number of folds). The chosen groups are refitted on all training data, projected as in Section 4.3 and instantiated as in Section 4.5. The ridge regularises ill-conditioned directions in standardised coordinates and favours small corrections, and along a direction the data do not determine it returns the nominal value; it does not establish identifiability.

Subset search is needed because aliased groups have different column spaces, so their residuals differ and, at low noise, the true group attains the minimum cross-validation score, whereas thresholding a group norm oscillates between aliased combinations and ends at the bounds (Appendix B). On the evaluated data, forward selection returns the same model as an exhaustive search over all subsets of at most four groups at a small fraction of its cost (Section 5.2).

Within a selected support, uniqueness holds under a rank condition.

Proposition 1 (Group-restricted uniqueness). *Let the discrepancy be confined to a known set of groups G , so that $r = A_G \xi_G^*$ holds exactly on the excitation data, and let A_G have full column rank. Then unregularised least squares on the columns of G returns ξ_G^* . If the rank falls short of the number of columns by k , a k -dimensional subspace of changes within G is not resolved by the data.*

The statement is the uniqueness of the normal equations, and it concerns the coefficients within a given support. Whether another support explains the same signal equally well is a different question, which depends on the union of the candidate column spaces. SAPC answers it empirically, by cross-validated selection and by repeating the selection over measurement-noise realisations, and it reports the answer as selection stability together with the empirical group-separation diagnostic of Definition 2.

4.5. Executable write-back

The identified increment becomes a simulator file, not a residual function. An inertial increment δ_b is added to π_b , projected if necessary and converted to the native fields of body b , mass, centre of mass, principal inertia and inertia frame, after which the simulator recomputes its derived constants. Coulomb and viscous increments are added to the joint's friction-loss and damping fields, whose totals the bounds keep non-negative, and an armature increment is added to its rotor-inertia field. The Stribeck, quadratic and offset terms have no native field in MuJoCo and are applied as resistive generalised forces, $\tau - \Phi(\dot{q})\xi$, with the same sign convention as the native friction loss and damping; they depend only on the current joint velocity and create no algebraic loop. Inertial changes are never applied as external forces: an inertial residual computed from a lagged acceleration closes an algebraic loop through the mass matrix and destabilises the twin.

The result is an executable calibrated twin that inherits the simulator's friction law, contact solver and integrator and runs downstream tasks the calibration never saw. Unlike a learned submodel or residual network [28, 27], the correction lives in the simulator's parameter space, where it can be read, checked, edited and transferred. With a reference generated by another simulator the same write-back serves the MuJoCo twin [32], and a physical term that

the reference description cannot express, such as a rotor inertia absent from an SDF model, is recovered as an armature increment (Section 5.7).

4.6. Why behavioral fidelity does not identify physics

The need for two validation axes follows from the structure of rigid-body dynamics. Let $\mathcal{N}$ be the subspace of inertial increments δ with $Y(q, \dot{q}, \ddot{q})\delta = 0$ at every state, where Y collects the regressors Y_b of all links.

Proposition 2 (Structural equivalence). *If two inertial parameter vectors differ by an element of $\mathcal{N}$, the corresponding twins have the same mass matrix $M(q)$ and bias force $h(q, \dot{q})$ at every state, and therefore the same inverse dynamics and, under the same actuation, friction model and numerical integration, the same free-space forward dynamics.*

Inverse dynamics is affine in the parameters, so $ID(\pi + \delta) = ID(\pi) + Y\delta = ID(\pi)$ for every $(q, \dot{q}, \ddot{q})$; it is affine in the acceleration, $ID = M(q)\ddot{q} + h(q, \dot{q})$ with M nonsingular, so equality for every $\ddot{q}$ gives $M = M'$ and $h = h'$, and $FD = M^{-1}(\tau - h)$ follows.

For the FR3, $\mathcal{N}$ has dimension 27, the difference between the 70 standard parameters and the 43 base combinations. An excitation that reaches this structural rank leaves no further null directions: two corrections that fit noise-free data exactly differ by an element of $\mathcal{N}$, and they are then indistinguishable in every free-space motion, not only on the visited states. Fitted corrections depart from this ideal through noise, regularisation, friction estimates and projection, so for them the proposition describes the dominant part of their difference, and Section 5.3 measures the rest. The identification trajectories of this study reach the structural rank (Section 5.4).

Attribution therefore requires a choice of representative within the equivalence class, that is, a prior. A minimum-norm dense fit chooses the representative closest to the nominal twin in the metric of its regulariser, a physically consistent dense fit chooses one that satisfies the consistency inequalities, and SAPC chooses the one supported on the fewest body and joint groups. The last choice is correct when the change is concentrated in groups whose torque signatures no other small set of groups reproduces, which is the usual situation of a twin that tracks a payload change, wear or a component replacement; Section 5.5 examines a distributed change and a concentrated one that another group reproduces. The representatives differ in use even though they cannot differ in free-space behavior: removing from each twin what it attributes to a component changes the representatives by different amounts, so their difference leaves $\mathcal{N}$, whereas subtracting the same known increment from both keeps them equivalent as long as the edited bodies remain physically valid (Section 5.3). Output-level credibility scores [11] and uncertainty analyses of twin outputs [30] cannot separate such twins; attribution, and its consequence under edits, can.

Contact transfer is not a corollary of Proposition 2. Contact geometry, the contact law and the constraint solver enter the dynamics separately from M and h , so twins with identical free-space dynamics can differ in contact, and across simulators they do (Section 5.7). That dense and structure-aware twins reproduce an unseen press task alike within one simulator (Section 5.6) is an observation consistent with the proposition, not a consequence of it; it is presented as downstream empirical validation.

5. Experiments

The experiments ask whether a calibration that reproduces behavior also attributes the discrepancy correctly, when a wrong attribution shows up in behavior, how stable the attribution is and how it degrades on other discrepancy patterns, and whether the answers hold when the reference is produced by Gazebo or Isaac Sim **and on a physical FR3 with a weighed payload**. Fig. 3 summarises the main comparison: in every row the nominal twin, the dense calibration and SAPC are compared on trajectory fidelity, torque fidelity, physical attribution and an unseen contact event.

5.1. Experimental design

All settings share one reference robot, one excitation family, one controller and one processing pipeline, so that the calibration schemes differ only in how they attribute the discrepancy.

Controlled same-simulator set-up Reference and twin are both the MuJoCo Menagerie FR3 model [33, 32], driven directly by generalised joint torques with contacts disabled in free space. The reference R carries the injected discrepancies of Table 1: a 0.8 kg payload body at the flange with an eccentric centre of mass, a Coulomb and a viscous friction increment on joint 4, a Coulomb increment on joint 6, and a 12 % increase of the mass and inertia of link 3. A payload change and a friction drift are the archetypal changes that a twin must track over the life of the

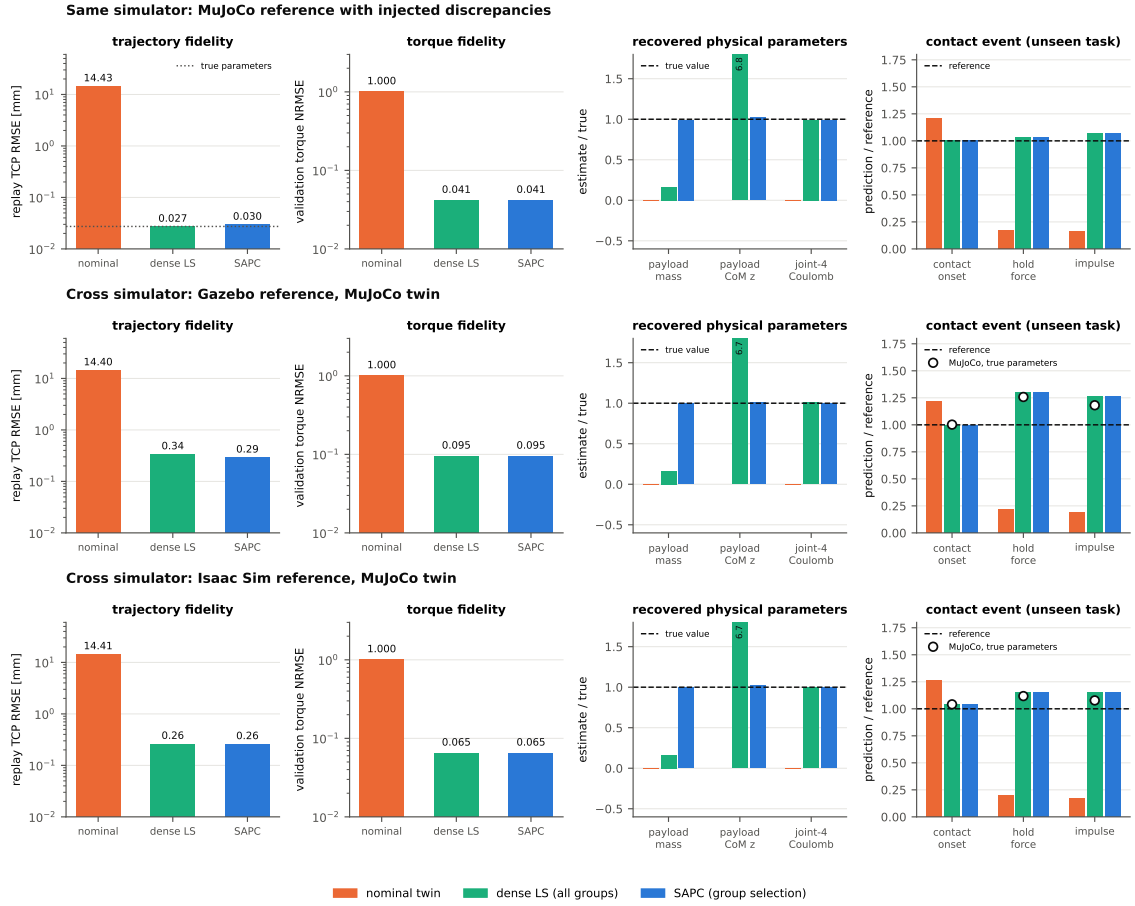


Figure 3: Same behavior, different physics. Columns compare the nominal twin, dense LS and SAPC against the reference on trajectory fidelity, torque fidelity, physical attribution and the unseen contact event. Rows: MuJoCo reference with the injected discrepancies; Gazebo reference; Isaac Sim reference; the twin is MuJoCo in every row. The first two panels of each row do not distinguish dense from structure-aware calibration, the third panel does, and the fourth shows that both calibrated twins predict the contact event alike. The dotted line in the first panel of the top row marks the twin with the true parameters; open markers in the contact panels of the cross-simulator rows give MuJoCo with the true parameters, i.e., the simulator-only contact discrepancy.

physical system [31], and injecting them at the model level provides an exact ground truth for physical attribution. The reference is integrated with MuJoCo’s implicit integrator at 0.5 ms; the twin T_0 is the nominal model with the implicit-fast integrator at 1 ms, and every calibrated twin inherits the twin’s settings. Instantiating the true parameters through the same write-back path gives a twin that replays the reference to a closed-loop TCP RMSE of 0.027 mm, against 14.43 mm for the nominal twin; this matched-parameter twin sets the scale of the numerical mismatch between the two configurations.

Excitation, noise and derivative estimation Excitation trajectories are five-harmonic Fourier series with a 0.1 Hz fundamental, following classical identification practice [15, 6]; each joint’s amplitude is scaled to 35 % of its range and capped at 1.6 rad/s and 8 rad/s². Each trajectory lasts 20 s; seeds 1-3 are used for identification and seeds 11-12 are held out for every behavioral metric. The reference tracks each trajectory with a nominal-feedforward PD controller, i.e., the inverse dynamics of T_0 plus a joint-space PD law, clipped at the FR3 torque limits of 87 N m (joints 1-4) and 12 N m (joints 5-7), so the residual torque is generated by the controller’s own effort to compensate the discrepancy. Joint positions and commanded torques are logged at 1 kHz. A sensor model adds encoder quantisation of 10^{-4} rad, Gaussian jitter of 10^{-5} rad and Gaussian torque noise of 0.05 N m, all multiplied by a common scale $s \in \{0, 1, 3\}$; $s = 1$ is the nominal level and $s = 3$ the stress level, and every training trajectory receives an independent noise stream.

Table 1

Injected physical discrepancies and candidate library.

Discrepancy	Reference value	Library group
Flange payload	0.8 kg at (0.02, -0.015, 0.06) m	link 7 (10 inertial parameters)
Joint-4 Coulomb / viscous friction	+1.66 N m / +0.34 N m s	joint 4 (5 friction terms)
Joint-6 Coulomb friction	+0.66 N m	joint 6
Link-3 mass and inertia	+12 %	link 3
Rotor inertia (Gazebo reference only)	absent	armature (7 terms)
Library	112 columns: 70 inertial, 35 friction, 7 armature	15 groups

Velocities and accelerations are estimated from the noisy positions with a Savitzky-Golay filter of 0.1 s window (101 samples) and polynomial order 3, the first and last 0.2 s of each trajectory are trimmed, and the rows are decimated to 200 Hz. Joint equations with $|\dot{q}_j| < 0.1$ rad/s are excluded from the identification rows, because the simulator's friction knee below this velocity is not in the library. Held-out trajectories are evaluated with the simulator's exact derivatives and torque noise at $s = 1$, and both torque metrics use all held-out samples, including the quarter of them below 0.1 rad/s; an exact-derivative condition on the identification data serves as the noise-free reference, and a learned derivative surrogate is compared with this pipeline in Appendix A.

Calibration schemes All schemes produce an executable calibrated twin through the same write-back path and use the same library and bounds unless stated otherwise. The *nominal twin* applies no correction. *Dense least squares* fits all 112 columns with a ridge of 10^{-6} in standardised units, which returns the minimum-norm correction; this is the dense baseline of the main text. Two dense variants isolate what the dense baseline lacks. The first uses exactly the fit of SAPC, bounded least squares with the ridge of 10^{-4} , on all groups at once, so that it differs from SAPC only in the absence of selection. The second additionally constrains the pseudo-inertia of every link to be positive definite inside the fit, $J(\pi_b + \delta_b) \geq 10^{-6} \mathbf{1}$, which yields physically consistent inertias without projection; it is our implementation of the linear-matrix-inequality estimators of [19, 7], solved with Clarabel through CVXPY. The *bounded elementwise sparse* baseline is STLSQ [21] with the box constraints and a threshold chosen by leave-one-trajectory-out cross-validation with the one-standard-error rule. SAPC is the method of Section 4, and *exhaustive best-subset* selection over all 1940 non-empty subsets of at most four of the 15 groups, plus the uncorrected twin, verifies its greedy search on the same data; the remaining schemes of the full benchmark are reported in Appendix B.

Metrics and additional tests Metrics follow Section 3. Behavioral fidelity is measured by the linear-library and executable-twin torque NRMSE, closed-loop replay TCP RMSE on the held-out trajectories and 0.2 s open-loop divergence; physical attribution by the payload mass and centre of mass, the link-3 mass increment, the joint-4 and joint-6 friction increments, and the precision and recall of the selected groups against the four injected groups. Three further tests probe the consequences and the limits of the attribution. An *editing test* changes each calibrated twin as a maintenance engineer would after a tool change: the link-7 increment, which a structure-aware twin attributes to the payload, is removed, or replaced by the known parameters of a new 0.5 kg tool, and the edited twin is compared with the reference that has undergone the same change; a control edits every twin with the known payload and tool inertias instead, and a transfer test without edit replays faster and larger motions than the identification data, with amplitudes of 45 % of the joint ranges, 2.5 rad/s and 12 rad/s². *Selection stability* repeats SAPC on the same reference trajectories over 20 sensor-noise realisations at $s = 1$ and 10 at $s = 3$, and group separation is diagnosed on the identification rows. *Other discrepancy patterns* are seven further references fixed before any fit (Section 5.5): six sparse cases with supports, sizes and locations different from the main study and one distributed case in which every link mass changes by 3-5 % in addition to the main discrepancies; each is calibrated from three noise realisations at $s = 1$ with all 15 groups as candidates.

Downstream contact task The executable calibrated twins are then used as test platforms [2] for an interaction that is absent from the calibration data: a probe-plate press. A spherical probe of 1 cm radius is attached at the flange, and a rigid plate is placed 26 mm below the probe bottom at the home pose, well below the sag of the loaded arm during settling. After 1 s of settling at home, a Jacobian-resolved joint reference drives the probe 29 mm downward over 1 s

with a cosine ramp, holds for 1 s and retracts over 1 s, under the same nominal-feedforward PD controller on every plant. The press is quasi-static, so it tests mainly the gravity load and joint friction of the twin at the press pose. The reference plate is stiffer and less damped than the plate the twins keep, and no contact parameter is fitted, so the twins differ only in the free-space calibration of the arm. The primary metrics are the contact onset (first sample whose normal force exceeds 0.05 N), the holding force (mean over 0.2-0.9 s of the hold), the impulse (integral of the normal force over the 4 s episode) and the TCP RMSE against the reference; the impact peak and the force-waveform RMSE are reported in Appendix D.

Cross-simulator references Simulators disagree with each other even at identical parameters [12], and simulator comparisons of manipulators have evaluated task outcomes rather than the underlying dynamics [13]. Two references produced by other simulators are therefore calibrated with the MuJoCo twin, which provides the simulator-native regressor and the write-back; the reference simulators are used only to generate observations. The first reference is Gazebo Harmonic. Its worlds are SDF files generated from the MJCF with the inertial parameters, joint damping and Coulomb friction copied, and a perturbed variant carries the discrepancies of Table 1 (Appendix C). SDF has no rotor-inertia field, so the reference has no rotor inertia, as deployed Gazebo FR3 models do; this description difference is not injected but inherited. The second reference is Isaac Sim 4.5 [34]. The same description is imported as a native fixed-base articulation that preserves the nominal rotor inertia, and the reference friction is applied as an explicit passive torque of the library's own form, because the Isaac Sim joint-friction parameter used here is a dimensionless coefficient rather than a torque; contact uses its rigid contact solver (Appendix C). The Gazebo world is stepped at 1 ms and the Isaac Sim scene at 0.5 ms; the controller applies the feedforward precomputed from the MuJoCo nominal inverse dynamics plus the same PD law at 1 kHz, and the derivative gain of joint 7 is lowered on every plant of these stages so that the 1 kHz discrete PD remains stable on a wrist joint without rotor inertia. Identification uses seeds 1-3 and validation seeds 11-12 of each perturbed reference, with the sensor model at $s = 1$ and Savitzky-Golay derivatives on both sets, since neither reference simulator exposes accelerations; cross-simulator torque metrics therefore include derivative-estimation error that the same-simulator validation, with exact derivatives, does not. Three levels are compared on the held-out trajectories: L1, the nominal reference against the nominal MuJoCo twin (simulator-only discrepancy); L2, the perturbed reference against the nominal twin; and L3, the perturbed reference against the calibrated twin, with the MuJoCo twin carrying the true parameters as the matched-parameter reference. Two controls accompany them. Calibrating the nominal twin on nominal reference data tests whether the method invents a physical discrepancy where none was injected, and a two-stage variant first absorbs whatever that calibration finds and then runs SAPC on the perturbed data against the twin calibrated on nominal reference data. For the Isaac reference, selection stability is measured on 10 training-noise realisations at $s = 1$ and 10 at $s = 3$ with the validation noise fixed at $s = 1$, and the no-injection control on 10 realisations of the nominal data. The press task is repeated in each perturbed reference world and compared with the MuJoCo twins on the nominal MuJoCo plate; MuJoCo with the true parameters provides the simulator-only contact reference.

5.2. Behavioral fidelity versus physical attribution

Table 2 and the top row of Fig. 3 give the within-simulator controlled validation. Every scheme that uses the full library, sparse or dense, explains the held-out residual equally well. The linear-library NRMSE is 0.041-0.042 at every noise level, and the executable-twin NRMSE, which evaluates the instantiated twin with the simulator's own friction law, is 0.024-0.025, the level of the torque-sensor noise. The library value is higher because of the library, not of the twins: for the true parameters, the relative residual is 0.026 on the measured torques of the rows above 0.1 rad/s, which is the sensor noise, 0.034 on noise-free torques when the slow rows are included, which is the friction knee, and 5×10^{-6} on noise-free rows above 0.1 rad/s. The replays agree as closely. At noise $\times 0$ and $\times 1$ the dense twins replay the held-out trajectories to 0.017-0.030 mm and SAPC to 0.028-0.030 mm, against 0.027 mm for the true parameters and 14.43 mm for the nominal twin. A validation protocol that stops at output-level agreement [9, 3] would accept any of these twins.

The attribution columns separate them. Dense least squares attributes the 0.8 kg flange payload to 0.13 kg and places its centre of mass 0.37 m from the true one, spreading the missing mass over links 3-6. Neither the fit nor the physical consistency of the dense baseline is the cause. The dense fit with SAPC's own bounded ridge routine returns 0.130 kg, and the fit that imposes the pseudo-inertia LMIs inside the optimisation, whose every body is realisable without projection, also returns 0.130 kg; the same holds at three times the nominal noise and with the Isaac Sim reference (Appendix B). The bounded elementwise sparse baseline shows that sparsity alone is not attribution: it replays to 0.03-0.09 mm while assigning 1.01-2.42 kg to the payload, because its support fragments across aliased

Table 2

Same-simulator calibration at three sensor-noise levels. Linear-library and executable-twin torque NRMSE on the held-out trajectories, closed-loop replay TCP RMSE, recovered payload mass (true 0.8 kg) and its centre-of-mass error, and group precision/recall. The dense variants use, in turn, a ridge of 10^{-6} ; SAPC's bounded fit with ridge 10^{-4} ; and the same fit with pseudo-inertia LMIs. The full comparison of all schemes is given in Appendix B.

Scheme	Noise	NRMSE lib.	NRMSE twin	TCP [mm]	Payload [kg]	CoM err. [mm]	Prec.	Rec.
Nominal twin	-	1.000	1.000	14.43	-	-	-	-
Dense LS	$\times 0$	0.042	0.024	0.017	0.130	368	0.27	1.00
Dense LS	$\times 1$	0.041	0.025	0.027	0.130	368	0.27	1.00
Dense LS	$\times 3$	0.042	0.025	0.067	0.131	368	0.27	1.00
Dense, SAPC fit	$\times 1$	0.041	0.025	0.028	0.130	369	0.27	1.00
Dense, pseudo-inertia LMI	$\times 1$	0.041	0.024	0.030	0.130	369	0.27	1.00
Bounded elementwise sparse	$\times 1$	0.042	0.025	0.035	1.112	67	0.50	1.00
Bounded elementwise sparse	$\times 3$	0.041	0.026	0.092	2.420	114	0.50	1.00
SAPC (proposed)	$\times 0$	0.042	0.024	0.028	0.795	1.1	1.00	1.00
SAPC (proposed)	$\times 1$	0.041	0.024	0.030	0.794	1.3	1.00	1.00
SAPC (proposed)	$\times 3$	0.042	0.025	0.062	0.779	5.1	0.75	0.75
Exhaustive best-subset (verification)	$\times 1$	0.041	0.024	0.030	0.794	1.3	1.00	1.00
True parameters	-	0.042	0.024	0.027	0.800	0.0	1.00	1.00

inertial columns. SAPC attributes the payload to 0.795 and 0.794 kg at noise $\times 0$ and $\times 1$ with a centre-of-mass error of about 1 mm, recovers the joint-4 Coulomb and viscous increments as 1.65 N m and 0.37 N m s (true 1.66 and 0.34), and selects exactly the four injected groups. At noise $\times 3$ link 4 takes the place of link 3 in this realisation and the payload estimate drops to 0.779 kg; Section 5.4 shows how often this happens and why. Forward selection returns the exhaustive optimum: on the same data, the search over all subsets of at most four groups returns the same support and parameters at every noise level and with exact derivatives, at about twenty times the cost (about 15 s against 5-6 min on one CPU core).

5.3. The consequence of attribution: editing the twin

Proposition 2 concerns the exact null directions; the fitted corrections differ from one another almost only along them, and the transfer test shows that the remainder is not visible in free space either. On faster and larger motions than any in the identification data, the dense twins replay to 0.022-0.026 mm and SAPC to 0.030 mm, against 0.025 mm for the true parameters and 13.5 mm for the nominal twin (Table 3). On generic random states the inertial torque of every fitted correction deviates from the true one by about 1 % of the injected torque, although the dense corrections lie five times farther from the truth in parameter space than SAPC's. Within the tested motions the twins behave as one dynamical system, not as one physical model.

The difference appears when the twin is edited, and how it appears depends on what the maintainer knows about the component. When a tool is taken off and the twin is edited through its component-wise corrections, without separate inertia information and without re-identification, the edit clears the identified link-7 increment (Fig. 4). This leaves the SAPC twin 0.10 mm from the unloaded reference, and replacing the increment by the known parameters of a new 0.5 kg tool gives 0.10 mm as well. The same edits leave 11.7 mm for every dense variant, the physically consistent one included, and 12.5 mm for the minimum-norm fit in raw units, worse than the uncalibrated twin at 2.6 mm, because the payload mass that the dense fits had placed on links 3-6 stays behind when the flange is cleared. At noise $\times 3$, where link 4 replaces link 3, the SAPC edit error grows to 0.36 mm while the dense ones stay at 11.7-12.5 mm. When the tool's inertia is known, the same known increment can be subtracted from every twin instead, and Proposition 2 predicts that the representatives stay equivalent. In the linear library they do: the known removal returns every twin to the torque NRMSE of the true parameters on the unloaded reference, 0.100-0.102, against 2.39-2.54 for the dense twins edited through their own attribution. The executable twin adds a constraint that the library does not have, the physical validity of every body. Subtracting the known 0.8 kg payload from the dense link 7, which carries only 0.13 kg of it, leaves that body with a negative mass of -0.04 to -0.08 kg; after the per-body pseudo-inertia projection of Section 4.3, the dense twins replay the unloaded reference to 1.2-1.7 mm, against 0.027 mm for SAPC, a result of this projection procedure rather than a bound for every way of restoring consistency. The exchange for the known 0.5 kg tool avoids negative

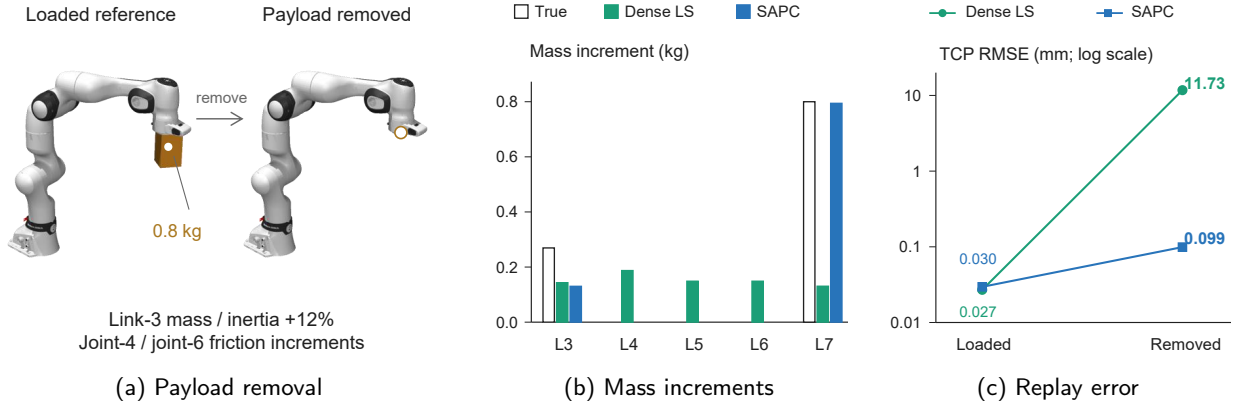


Figure 4: Physical attribution under a payload edit made through each twin’s own attribution, noise $\times 1$, held-out trajectories. (a) The 0.8 kg payload is removed while the link-3 mass and inertia increase and the joint-4 and joint-6 friction increments remain; each twin is edited by clearing its link-7 increment, without refitting, and the amber block only illustrates the payload’s mass, centre of mass and principal moments. (b) True and identified mass increments per link; links 1 and 2 carry none in any model, and because the link-3 mass is structurally unidentifiable, the smaller SAPC increment there implies no error in its torque signature. (c) Replay TCP RMSE before and after the edit on a logarithmic scale; the uncalibrated twin reaches 2.60 mm after removal.

Table 3

Editing test at noise $\times 1$. Identified edit: the twin’s own link-7 increment is removed or replaced by the known parameters of a new 0.5 kg tool. Known-inertia edit: the known payload increment is subtracted, and for the new tool its known parameters added, whatever the twin attributed to link 7; a link that becomes physically invalid is projected as in Section 4.3. Each edited twin is replayed against the reference after the same physical change; the last column replays faster and larger motions without any edit. For the nominal twin the edits add or remove only the known tool. Links 1-6: sum of the absolute mass increments of those links, whose true value is the 12 % increase of link 3.

Twin	Mass increment [kg]		Identified edit [mm]		Known-inertia edit [mm]		Faster motions [mm]
	Link 7	Links 1-6	Removed	New tool	Removed	New tool	
Nominal twin	0	0	2.60	2.61	2.60	2.61	13.49
SAPC	0.794	0.130	0.099	0.100	0.027	0.029	0.030
Dense LS	0.130	0.628	11.73	11.73	1.20	0.030	0.022
Dense, SAPC fit	0.130	0.628	11.74	11.74	1.20	0.030	0.024
Dense, pseudo-inertia LMI	0.130	0.628	11.74	11.73	1.20	0.031	0.026
Dense, raw-unit minimum norm	0.090	0.243	12.45	12.45	1.68	0.037	0.023
True parameters	0.800	0.269	0.026	0.027	0.026	0.027	0.025

link masses and requires only small consistency corrections, a relative projection of 0.6-0.9 % of the dense link 7, and every twin replays to 0.029-0.037 mm (Table 3). Correct attribution is therefore required when a twin is edited through its component-wise corrections without separate inertia information or re-identification, and with the projection used here a known removal also stays accurate only when the attribution leaves the edited body physically valid.

5.4. Stability and identifiability

Fig. 5 and Table 4 report SAPC over repeated sensor-noise realisations on the same reference trajectories. At noise $\times 1$ the four injected groups are selected in all 20 MuJoCo and all 10 Isaac Sim realisations, and no other group in any. The payload mass is 0.794 ± 0.001 kg in MuJoCo and 0.795 ± 0.001 kg in Isaac Sim, and the joint-4 Coulomb increment 1.653 ± 0.003 and 1.655 ± 0.002 N m (true 1.663). At noise $\times 3$ the payload and both friction groups are still selected in every realisation, whereas link 3 is selected in 8 of 10 MuJoCo and 7 of 10 Isaac Sim realisations and link 4 takes its place in the others. Precision degrades with the substitution: in Isaac Sim the three link-4 realisations have a mean payload of 0.776 kg and a centre-of-mass error of 6.0 mm, against 0.793 kg and 1.6 mm when link 3 is kept.

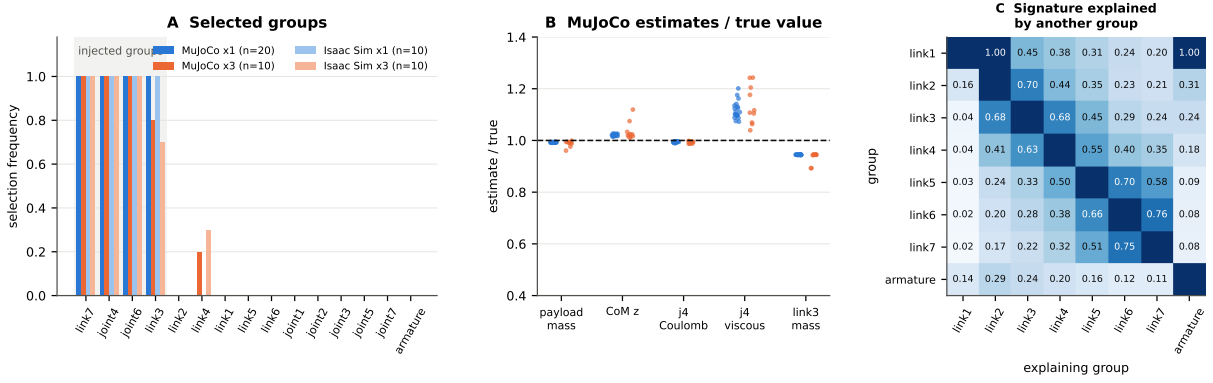


Figure 5: Selection stability and group separation. (A) Selection frequency of each group over 20 (MuJoCo, noise $\times 1$), 10 (MuJoCo, $\times 3$) and 10 (Isaac Sim, $\times 1$ and $\times 3$) sensor-noise realisations. (B) MuJoCo estimates normalised by the true value; the link-3 mass is a structurally unidentified direction. (C) Fraction of a group's standardised torque signature explained by another group's column space on the identification rows.

Table 4

Selection stability of SAPC over sensor-noise realisations: selection counts of the injected groups, of the aliased neighbour link 4 and of the empty model, and estimates at noise $\times 1$ (mean $\pm$ standard deviation, true value in parentheses). No other group is selected in any realisation.

Reference, noise	n	link 7	joint 4	joint 6	link 3	link 4	none
MuJoCo, $\times 1$	20	20/20	20/20	20/20	20/20	0/20	0/20
MuJoCo, $\times 3$	10	10/10	10/10	10/10	8/10	2/10	0/10
Isaac Sim, $\times 1$	10	10/10	10/10	10/10	10/10	0/10	0/10
Isaac Sim, $\times 3$	10	10/10	10/10	10/10	7/10	3/10	0/10
Isaac Sim nominal, $\times 1$ (control)	10	0/10	0/10	0/10	0/10	0/10	10/10
<i>Estimates at noise $\times 1$</i>							
Payload mass [kg]	MuJoCo 0.794 ± 0.001 (0.800)			Isaac Sim 0.795 ± 0.001 (0.800)			
Payload CoM error [mm]	MuJoCo 1.3 ± 0.2			Isaac Sim 1.1 ± 0.1			
Joint-4 Coulomb [N m]	MuJoCo 1.653 ± 0.003 (1.663)			Isaac Sim 1.655 ± 0.002 (1.663)			
Control payload [kg]	Isaac Sim nominal: empty support in 10 of 10 realisations, no correction of any group						

The structural audit explains the pattern. Of the 70 inertial parameters, 43 combinations affect the torques, and the identification excitation reaches all 43 with a condition number of 35 over the base directions, so the aliasing is not an excitation deficit. Neighbouring links share structural subspaces, two dimensions between links 2 and 3 and three between each later pair, and the injected link-3 change can be reproduced to 99.1 % of its torque energy by link 4 alone and to 99.997 % by links 2 and 4 together; the substitution at high noise is a choice between nearly equivalent sparse explanations. The payload's torque is 99.6 % reproducible by a change of link 6, because the payload sits close to the axis of joint 7, where a point mass does not move with that joint; its attribution to the flange rests on the lateral offset and the inertia that link 6 cannot represent, and cross-validation resolves this remainder in every realisation. The mass of link 3 on its own is structurally unidentifiable. A mass at the intersection of the first three joint axes, 0.316 m from the link-3 origin, produces no torque in any motion, 3×10^{-14} N m per kg against 0.81 N m per kg for a uniform scaling of link 3, and it is the single null direction of the link-3 columns. SAPC leaves this direction at its nominal value, which is why the reported link-3 mass increment, 0.13 kg, differs from the injected 0.27 kg although the torque effect of the correction is right.

The no-injection control completes the picture. Calibrated on nominal Isaac Sim data, SAPC returns the uncorrected twin in all 10 realisations: the empty model lies within one standard error of the best single group, the payload increment is zero, the torque NRMSE is 1.000 and the replay error stays at the simulator-only 0.19 mm. Without the empty model among the candidates, each realisation had selected one negligible group; admitting it is what lets the method report that nothing changed. Selection frequencies, estimate spreads and structural diagnostics thus attach the uncertainty of a calibration to named bodies and joints, rather than to the outputs [30] or to one aggregated score [11].

Table 5

Other discrepancy patterns, three noise draws at $\times 1$ each. Selection: groups chosen by SAPC and in how many draws. Estimate: SAPC's increments (true values in parentheses). Dense payload: payload mass attributed by the dense LMI fit. Replay: SAPC and true-parameter replay TCP RMSE.

Case (injected groups)	Selection	Estimate	Dense payload [kg]	Replay [mm]
C1 payload 1.2 kg (link 7)	link 7 (3/3)	payload 1.200 kg (1.2), CoM 0.1 mm	0.19	0.025 / 0.028
C2 friction (joint 2)	joint 2 (3/3)	joint 2: 1.50 N m, 0.39 N m s (1.5, 0.4)	-	0.025 / 0.026
C3 mass +25 % (link 5)	link 5 (3/3)	link 5: 0.581 kg (0.582)	-	0.021 / 0.027
C4 payload 0.4 kg (link 7, joints 1, 7)	joint 1, joint 7, link 7 (3/3)	payload 0.400 kg (0.4), CoM 0.3 mm; joints 1, 7: 2.00, 0.30 N m (2.0, 0.3)	0.06	0.024 / 0.027
C5 payload on joint-7 axis (link 7, link 6, joint 3)	joint 3, link 6 (3/3)	link 6: 1.36 kg (0.36, plus the 1.0 kg payload on link 7)	0.22	0.024 / 0.029
C6 armature, joint 4, link 2	armature, joint 4, link 2 (2/3); armature, joint 4, link 3 (1/3)	armature j5 0.051 (0.050); joint 4 1.00 N m (1.0)	-	0.024 / 0.025
C7 distributed (all links, joints 4, 6)	joint 4, joint 6, link 2, link 7 (3/3)	payload 0.750 kg (0.8), CoM 7.0 mm; joints 4, 6: 1.66, 0.66 N m (1.66, 0.66)	0.12	0.036 / 0.027

5.5. Other discrepancy patterns

Table 5 applies the unchanged pipeline to seven further references, fixed before any fit, with all 15 groups as candidates. In four of the six sparse cases SAPC selects exactly the injected groups in all three noise draws and recovers the magnitudes closely: a 1.2 kg payload at another location to 1.200 kg with a centre-of-mass error of 0.1 mm, a friction change on joint 2, a 25 % mass increase of link 5 to 0.581 kg against 0.582 kg, and a light payload combined with friction changes on joints 1 and 7. In the armature case, link 3 replaces link 2 in one of three draws, the aliasing of Section 5.4. Case C5 is a structural limit that the audit predicts. Its payload sits on the axis of joint 7, where a point mass is indistinguishable from a mass change of link 6 in any motion except through its small inertia about the axis, so SAPC attributes the payload and the link-6 change to link 6 alone, which reproduces the torques with one group instead of two. The dense fits, as always, spread the payload over all links. In the distributed case C7, which violates the sparse-change prior, SAPC still recovers the dominant changes, the payload within 7 % and 7 mm and the friction increments exactly, and represents the 3-5 % mass changes of all links by a single link group. Its replay error stays close to those of the true parameters and of the dense fits, which attribute 0.12 kg to the payload. The prior thus degrades gracefully when it is mildly violated: small distributed changes are absorbed, and the large concentrated ones are still attributed.

5.6. Out-of-sample contact behavior

Table 6 and the fourth panel of Fig. 3 evaluate the twins on a task that calibration never saw, a probe-plate press with no contact parameter fitted. The nominal twin misses the event itself: contact starts 0.31 s late, the holding force is 5.7 times too small and the impulse 6 times too small. All calibrated twins recover the event from free-space data alone, the onset within 1 ms, the holding force within 3 % and the impulse within 7 %, with the TCP error reduced from 10.8 mm to 0.72-0.73 mm; the true parameters with a matched integrator reproduce the reference to 0.018 mm. The calibrated rows agree to within 0.01 N and 0.01 N s, although one carries a 0.13 kg payload at the wrong place and another a 0.79 kg payload at the right place. The press is quasi-static and tests mainly the gravity load at one pose, which every torque-equivalent correction reproduces. Out-of-sample contact success therefore does not establish physical correctness; the editing test does separate the twins.

5.7. Cross-simulator calibration and discrepancy decomposition

Table 7 and the lower rows of Fig. 3 repeat the experiment with references generated by Gazebo and by Isaac Sim while the twin stays in MuJoCo, and Fig. 6 decomposes the replay error.

Table 6

Unseen press task (26 mm plate gap, 29 mm descent), same simulator. Calibration used free-space data only; no contact parameter is fitted.

Twin	Onset [s]	Holding force [N]	Impulse [Ns]	TCP RMSE [mm]
Reference	1.485	10.59	17.46	-
Nominal twin	1.797	1.85	2.79	10.80
Dense LS calibrated	1.486	10.90	18.63	0.72
Dense, pseudo-inertia LMI calibrated	1.486	10.90	18.63	0.73
SAPC calibrated	1.486	10.90	18.63	0.73
True parameters, matched integrator	1.485	10.50	17.20	0.018

Gazebo At L1, nominal Gazebo against nominal MuJoCo, the replay TCP RMSE is 0.52 mm, the disagreement the two simulators produce on their own; at L2, the perturbed reference against the nominal twin, it is 14.40 mm. At L3 dense calibration brings it to 0.34 mm and SAPC to 0.29 mm at the same validation torque NRMSE of 0.095, and the same-simulator pattern repeats: dense calibration reports the payload as 0.131 kg spread over all 15 groups, whereas SAPC reports 0.797 kg at (0.0199, -0.0153, 0.0606) m, within 0.4 % in mass and 1 mm in centre of mass. The torque NRMSE is 2.3 times that within one simulator because it also contains the error of the derivatives estimated from Gazebo data and Gazebo's friction and constraint behavior, which the library does not represent. SAPC additionally finds a discrepancy that was never injected. The SDF description has no rotor inertia, whereas the MuJoCo nominal model carries an armature of 0.195 kg m^2 on joints 1-4 and 0.074 kg m^2 on joints 5-7, a representable parametric discrepancy between the two simulator descriptions. The armature group is selected with corrections of -0.19 and -0.18 on joints 3-4 and -0.07 on joints 5-7, matching the missing values, and smaller corrections on joints 1-2, where the link inertia dominates the rotor term; two further groups, link 2 and joint 7, enter as compensation for simulator-specific friction and stiction behavior. Calibrating on nominal Gazebo data alone selects exactly the armature group and joint 7, recovers the armature on all seven joints (-0.194 to -0.190 on joints 1-4, -0.070 to -0.074 on joints 5-7), assigns no mass to the payload and reduces the simulator-only replay error from 0.52 to 0.11 mm (Fig. 6). Most of the raw cross-simulator discrepancy is thus a representable description difference. Removing the armature from the calibrated twin worsens its replay to 0.54-0.58 mm, above the simulator-only level, whereas the injected groups plus the armature replay at 0.20 mm. The two-stage variant turns the decomposition into a procedure: stage one calibrates the nominal twin on nominal Gazebo data and absorbs the armature and joint 7; stage two runs SAPC against that twin and selects the injected groups and joint 7, dropping link 2, with a replay error of 0.20 mm and a payload of 0.798 kg, while dense calibration on the same stage-one twin still returns 0.131 kg.

Isaac Sim The Isaac Sim reference preserves the rotor inertia, so it differs from the twin in its solver, its contact model, its friction law at low speed and the injected discrepancies. At L1 the simulator-only replay discrepancy is 0.19 mm, the matched-parameter twin replays the perturbed reference to 0.26 mm, and at L2 the nominal twin deviates by 14.41 mm. At L3 dense calibration and SAPC both reach 0.26 mm at a validation torque NRMSE of 0.065, the level of the matched-parameter twin, and once more they part on attribution: dense calibration reports 0.131 kg over all 15 groups, SAPC reports 0.795 kg at (0.0202, -0.0148, 0.0611) m with exactly the four injected groups, no armature correction and no compensating group. Calibrated on nominal Isaac Sim data, SAPC returns the uncorrected twin, and the two-stage variant therefore coincides with the single-stage calibration. Where the simulator descriptions agree, the calibration finds only what was injected; where they differ, as with the rotor inertia absent from SDF, it finds the difference and names it.

Contact transfer Each reference presses on its own plate and the MuJoCo twins on the nominal MuJoCo plate. Against Gazebo, the calibrated twins reach the plate within 1 ms of the reference and overshoot its holding force and impulse by factors of 1.30 and 1.26, while MuJoCo with the true robot parameters overshoots them by 1.26 and 1.18. Against Isaac Sim, the calibrated twins reach the plate 60 ms after the reference and hold 1.15 times its force, while MuJoCo with the true parameters shows the same delay and 1.12 times the force. Most of the contact residual is therefore the difference between the contact laws and solvers of the simulators at identical robot parameters, which no parameter in the library represents, and as within one simulator the dense and structure-aware twins are indistinguishable on the contact metrics.

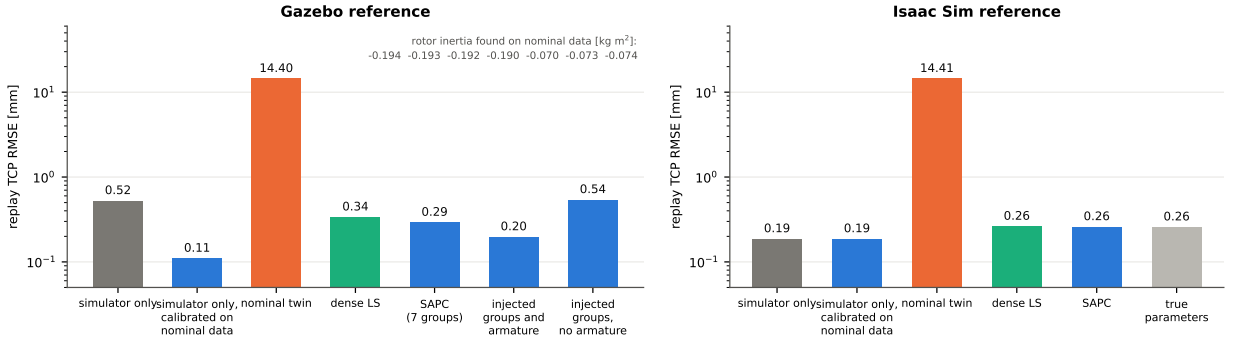


Figure 6: Cross-simulator decomposition of the replay TCP RMSE. Left, Gazebo: calibrating on nominal Gazebo data alone discovers the missing rotor inertia of all seven joints and reduces the simulator-only discrepancy from 0.52 to 0.11 mm; removing the armature from the calibrated twin raises the error above the simulator-only level. Right, Isaac Sim: the rotor inertia is preserved, calibration on nominal data returns no correction, and the calibrated twins reach the matched-parameter twin.

Table 7

Cross-simulator calibration with the MuJoCo twin. Groups, validation torque NRMSE (linear library, SG derivatives), closed-loop replay TCP RMSE on the held-out trajectories, recovered payload mass (true 0.8 kg) and holding force in the press task.

Level / twin	Groups	NRMSE	TCP [mm]	Payload [kg]	Hold [N]
<i>Gazebo reference (holding force 8.42 N)</i>					
L1 simulator only (nominal vs nominal)	-	-	0.52	-	10.59 [†]
L1 SAPC on nominal data	armature, j7	0.41	0.11	0.000	-
L2 nominal twin	-	1.000	14.40	-	1.85
L3 dense LS	all 15	0.095	0.34	0.131	10.96
L3 SAPC	7	0.095	0.29	0.797	10.95
L3 injected groups + armature	5	0.095	0.20	0.794	-
L3 injected groups, no armature	4	0.100	0.54	0.784	-
L3 two-stage SAPC	5	0.104 [‡]	0.20	0.798	-
<i>Isaac Sim reference (holding force 9.44 N)</i>					
L1 simulator only (nominal vs nominal)	-	-	0.19	-	10.55 [†]
L1 SAPC on nominal data (control)	none	1.000	0.19	0.000	-
L2 nominal twin	-	1.000	14.41	-	1.85
L3 dense LS	all 15	0.065	0.26	0.131	10.90
L3 SAPC	4 (injected)	0.065	0.26	0.795	10.90
L3 two-stage SAPC	4 (injected)	0.065	0.26	0.795	-
True parameters (Isaac Sim reference)	-	-	0.26	0.800	10.55

[†]MuJoCo with the true robot parameters against the same reference press. [‡]Relative to the stage-one residual.

5.8. Derivative-estimation and residual-model comparisons

Two learned alternatives to the structured pipeline were tested; Appendix A gives the details. A derivative surrogate adapted from physics-informed sparse regression [22], a per-trajectory Fourier-plus-MLP fit of $q(t)$ with automatic differentiation and alternating physics refinement, follows the route that physics-informed surrogates trained on simulator data take for contact-rich dynamics [26]. Under the 1 kHz band-limited excitation used here its velocity error was about 20 times and its acceleration error about 6 times that of the tuned Savitzky-Golay estimate, and identification by SAPC on the surrogate derivatives reached a validation NRMSE of 0.061 and a replay error of 0.46 mm, against 0.042 and 0.062 mm with Savitzky-Golay derivatives on the same data. A forward-residual MLP [27], the black-box alternative to a parametric correction, fits the training trajectories but reaches a validation NRMSE of 0.60-0.72 and leaves the replay error at 13.9-16.7 mm, so it does not improve the nominal twin. Both results agree with the hybrid-twin

Table 8

Payload attribution on a physical FR3, relative to the stage-one twin of the unloaded robot: torque RMSE on the held-out trajectory with the block, and on the second unloaded run after the identified link-7 increment is removed, which for the stage-one twin is its own prediction.

Twin	Stage-two groups	Link-7 mass [kg]	CoM err. [mm]	Loaded [N m]	Unloaded, payload removed [N m]
Stage-one twin	-	-	-	1.52	0.41
Dense LS	all 15	0.19	310	0.42	0.94
SAPC	link 7	0.791	3.8	0.42	0.41
Weighed block	-	0.812	0.0	-	-

practice of coupling a simulator with a learned submodel only where the submodel is trusted [28]: for a smooth, well-excited manipulator the simulator-native regressor is the trusted part, and these comparisons select the Savitzky-Golay derivatives for this data regime without being a general verdict on physics-informed surrogates.

5.9. Payload attribution on a physical FR3

A physical robot offers one change whose ground truth is exact, a payload whose mass is weighed. A steel block of 0.812 kg, weighed to 1 g, was bolted to the flange of an FR3 without its hand, with its centre of mass at (0.021, -0.014, 0.058) m in the flange frame according to its CAD model; the offset from the joint-7 axis keeps link 7 separable from link 6, unlike case C5 of Section 5.5. Three excitation trajectories as in Section 5.1, at 60 % of its velocity and acceleration caps and with 2 s ramps from and to rest, two for identification and one held out, were tracked by the robot's joint impedance controller without the block, with it and again without it, recording joint positions and link-side joint torques at 1 kHz as in [6]. Calibration follows the two-stage protocol of Section 5.7: the first unloaded run calibrates the nominal twin to the robot, without the eight-group cap of Section 4.4 because a link-side torque differs from the actuator torque of the twin at every joint, and the loaded run identifies the change relative to the complete stage-one twin. Stage one selects nine groups, among them the armature, which link-side torque sensors do not see: the group absorbs the acceleration-proportional part of the difference between the link-side torque and the actuator torque of the twin, which no inertial column of a single link represents. The stage-one twin predicts the held-out unloaded trajectory to a torque RMSE of 0.40 N m before the loaded runs and 0.41 N m after them, so the robot did not drift between the runs.

Stage two selects link 7 alone and attributes 0.791 kg to it, 2.6 % below the weighed mass, with a centre-of-mass error of 3.8 mm (Table 8). Dense calibration of the same change predicts the held-out loaded trajectory equally well, to 0.42 N m, but assigns 0.19 kg to link 7 and 0.57 kg of absolute mass increments to the other links. With the identified link-7 increment removed, the SAPC twin predicts the second unloaded run to 0.41 N m, the value of the stage-one twin, whereas the dense twin misses it by 0.94 N m. On the physical robot as in simulation, equal fidelity on the loaded robot hides a different attribution, and only the twin that attributes the change correctly remains valid after the edit.

6. Discussion

The results carry one message: a robot digital twin can reproduce measured, held-out, extrapolated and unseen contact behavior while containing the wrong physics, and the difference becomes visible, and costly, when the twin is used as a model of the physical system rather than as a replay of it. We draw out what this implies for the meaning of calibration, for the role of priors and identifiability, for the practical value of structure and for the reading of cross-simulator residuals, and we state the limits of the study.

6.1. What calibrated means

A twin should not be declared physically correct because it matches training torque, held-out or faster trajectories, or an unseen contact event. In the controlled study the dense and structure-aware twins agree on all of these to within the numerical mismatch of the matched-parameter twin, because their corrections differ almost only along structurally null directions (Proposition 2), and they separate when a tool is removed through the twin's own attribution, the most ordinary edit in service. We therefore recommend reporting three quantities separately: behavioral fidelity, with the library and executable-twin torque metrics, replay and contact metrics; physical attribution, with the selected groups,

their increments and their selection stability; and the residual model-form discrepancy, measured by a twin that carries the reference's true parameters where they are known. When the twin is maintained by editing it, an edit test of the kind in Section 5.3 belongs in the validation as well. Credibility frameworks ask for evidence proportional to the decision a model informs [8]; for a twin that is used to decide which component changed or to predict the effect of an exchange, attribution evidence is that evidence, and output-level validation [9, 10, 11] does not provide it.

6.2. Attribution needs a prior, and identifiability sets its resolution

Behavior identifies the base-parameter class of a correction, and attribution within the class is a choice. A minimum-norm dense fit makes that choice through the metric of its regulariser, so its payload estimate changes with the metric while its torque fit does not; a physically consistent dense fit restricts the choice to realisable bodies, which the true correction and the dense one both are; SAPC makes it through the sparse-change prior, which is right when the change is concentrated in groups that the data can tell apart. Within that prior the data still set a resolution, which the empirical group-separation diagnostic and the selection stability report. The payload's torque signature is almost entirely reproducible by a change of link 6, and its attribution to the flange rests on the small remainder, which cross-validation detects in every realisation at both noise levels and in both simulators. The injected link-3 change is even closer to a combination of links 2 and 4, and at three times the nominal noise link 4 takes its place in a minority of realisations, with larger payload and centre-of-mass errors when it does; the selection frequency reports this instead of hiding it. When the prior is violated, the attribution degrades gracefully: in the distributed case SAPC recovers the payload to 0.75 kg and the joint friction increments almost exactly, while the 3-5 % mass changes of the other links are represented by a single link group. Admitting the uncorrected twin as a candidate completes the picture: where nothing was injected, SAPC returns no correction, whereas a selection path that starts at the first group always returns one.

6.3. Why structure matters

Dense regression solves the behavioral fitting problem; structure-aware calibration also answers which physically interpretable subsystem changed, and the editing test shows that the answer has a behavioral value, not only an interpretive one. The same holds for any maintenance action that is expressed as an edit of the twin's own corrections: a tool change edits the flange group and leaves the friction corrections in place, a joint service resets one friction group, and a component replacement changes one body. Writing the correction into the native fields of a rigid-body description also makes it transferable across simulators, and it is why the rotor inertia missing from the SDF reference is identified as a rotor-inertia deficit rather than absorbed into link inertias. Twins used as test platforms for planning and control [2] inherit whatever physics they contain, and residual learning [27, 24] or monolithic parameter tuning [4, 5] gives behavioral fidelity without a physical address for the correction.

6.4. Cross-simulator interpretation

A cross-simulator residual has two parts: representable parametric discrepancy, which the calibration library can express and SAPC identifies, and residual model-form discrepancy, which the library cannot express. The two reference simulators illustrate both. Most of the simulator-only replay discrepancy between Gazebo and MuJoCo is the rotor inertia that the SDF description cannot carry, and calibration on nominal Gazebo data alone recovers it on all seven joints with no spurious payload; removing the armature group from the calibrated twin raises the replay error above the simulator-only level. Isaac Sim preserves the rotor inertia, and there the calibration on nominal data finds nothing to correct and the perturbed calibration recovers exactly the injected groups. What remains after calibration is measured against the matched-parameter twin, not against a universal floor: it contains the difference between Gazebo's rigid stiction and MuJoCo's soft friction constraint [32, 35], the different articulation formulation and constraint stabilisation of Isaac Sim, and, in the torque metrics, the derivative estimation that both cross-simulator validations require. Output-level simulator comparisons [12, 3] record such disagreements without attributing them; the decomposition says which part of a disagreement is a parameter and which is a model.

6.5. Limitations

The scope of the study is set by limits of three kinds, and each comes with a way to address it. Some hold for any calibration from free-space joint torques: the link-3 mass is a structurally null direction, and a payload on the joint-7 axis, as in case C5, produces the torques of a link-6 change; only information beyond the joint torques, such as CAD data or a base force-torque sensor, resolves them. Others are detected by the diagnostics of the method. Near-aliased link groups can swap under strong measurement noise, which the selection frequency reports and excitation designed

on the group-separation diagnostic can prevent. A distributed change such as case C7 is recovered in its concentrated part while its small mass changes are merged into one group, and grouping the bodies that a maintenance action affects together matches the prior to the action. Selection stability resamples measurement noise on fixed trajectories and is not a probability of being correct; resampling the trajectories as well extends it to the variability of the excitation. The remaining limits mark next steps. The press does not identify contact stiffness or damping, and its force-waveform metrics are solver and time-step sensitive (Appendix D), so a contact-identification step with a force sensor can follow the free-space calibration once the rigid-body part is fixed. The cross-simulator residuals are specific to the 112-column library, and the Isaac Sim reference applies its friction in the library's own form, so a reference with a friction law outside the library would test that case. Forward selection agrees with the exhaustive search over at most four groups on the evaluated data, and mixed-integer best-subset formulations can certify the best subset of each size for larger supports.

7. Conclusion

A robot digital twin answers two questions, whether it behaves like the reference system and whether it contains the right physics, and on a 7-DoF manipulator with exactly known injected discrepancies the answers differ. Dense re-identification, with or without physically consistent inertias, and SAPC reproduce held-out and faster motion and an unseen press task alike, yet the dense twins attribute the injected 0.8 kg payload to 0.13 kg spread over neighbouring links whereas SAPC recovers 0.79 kg at the flange. The corrections differ almost only along structurally null directions, and none of the tested free-space motions separates them; when the payload is removed through each twin's own attribution, without separate inertia information or re-identification, the dense twins miss the unloaded reference by 11.7-12.5 mm against 0.10 mm for SAPC, worse than no calibration for every dense variant. Subtracting the separately known payload instead restores every twin in the linear model, but in the simulator it leaves the dense flange body with a negative mass and, after the per-body projection used here, a replay error of 1.2-1.7 mm against 0.027 mm for SAPC. SAPC obtains this attribution with a simulator-native regressor, bounded forward selection over body and joint groups that admits the uncorrected twin, and write-back into an executable calibrated twin; its selection stability separates robust attributions from near-aliased ones, and it returns no correction when nothing changed. Cross-simulator validation against Gazebo and Isaac Sim references repeats the pattern in both simulators, recovers the rotor inertia missing from the SDF description from nominal data alone, and leaves in each case a residual measured against the twin with the true parameters. **On a physical FR3, the two-stage protocol attributes a weighed 0.81 kg payload to the flange within 3 %, where dense calibration places 0.19 kg, and after its removal only the SAPC twin predicts the unloaded robot as well as the twin calibrated on it.** Validation of a digital twin should therefore report behavioral fidelity, physical attribution and residual model-form discrepancy as three separate quantities, and it should test the twin under the edits it will undergo in service.

CRedit authorship contribution statement

Suhwan Park: Methodology, Software, Validation, Investigation, Writing - original draft. **Jihwan Lee:** Investigation, Validation. **Jiyong Park:** Investigation, Validation. **Sanghyun Kim:** Conceptualization, Methodology, Supervision, Funding acquisition, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Every result in this paper can be reproduced from public data. The project page, <https://rcilab.khu.ac.kr/sapc>, provides three archives: the result files behind every table and figure; the recorded Gazebo and Isaac Sim reference logs for identification, validation and the press task, **together with the joint logs of the physical FR3**; and the robot models and generated simulator worlds. It also provides the source code of SAPC and of every experiment, including the MJCF-to-SDF and MJCF-to-USD conversions, the simulator bridges and the fixed-seed scripts that regenerate the MuJoCo trajectories.

A. Learned derivative surrogates

This appendix documents the learned derivative surrogate compared with Savitzky-Golay (SG) differentiation. Physics-informed surrogates trained on simulator data have been proposed for contact-rich robot dynamics [26], and physics-informed sparse regression [22, 23] is the natural candidate for replacing filtered derivatives in a sparse identification. Our adaptation fits one network per trajectory, $\hat{q}(t) = q_0 + W\phi(t) + \text{MLP}(\phi(t))$, with Fourier features ϕ up to $f_{\max} = 2.5$ Hz, a least-squares-initialised linear part and a two-layer MLP (64 units, zero-initialised output) that adds only what the Fourier basis cannot represent; derivatives come from automatic differentiation. Pre-training adds a smoothing penalty on $\hat{q}$ whose weight is set by the discrepancy principle, the largest value on the grid $\{0.1, \dots, 1000\}$ that keeps the per-sample χ^2 below 1.3. Alternating direction optimisation then repeats, for two rounds, a bounded sparse fit on the surrogate derivatives and a refinement of the network under the data loss plus the physics loss $\|\tau - \text{ID}_c(\hat{q}, \dot{\hat{q}}, \ddot{\hat{q}}; \xi)\|^2 / \sigma_\tau^2$, where ID_c is the inverse dynamics of the calibrated simulator model and its Jacobians with respect to $(q, \dot{q}, \ddot{q})$ come from MuJoCo's finite-difference routine `mjd_inverseFD` through a custom autograd function. This is an adaptation for the present data regime, not a reproduction of the original method.

At noise $\times 3$ on the three 20 s training trajectories the surrogate derivatives were less accurate than SG: RMS errors of 0.0154-0.0189 rad/s in $\dot{q}$ and 0.1834-0.2310 rad/s² in $\ddot{q}$, against 0.0008-0.0009 rad/s and 0.0310-0.0357 rad/s² for SG. The training torque NRMSE of the sparse fit on the surrogate derivatives was 0.091 after pre-training (57 non-zeros) and 0.249 (53 non-zeros) and 0.239 (69 non-zeros) after the refinement rounds, because the physics loss pulls the surrogate towards a still incomplete sparse model. The identification inherits the loss: on the same data, bounded elementwise sparse regression reached a validation torque NRMSE of 0.041 and a replay error of 0.092 mm with SG derivatives against 0.059 and 0.47 mm with the surrogate, and SAPC reached 0.042 and 0.062 mm against 0.061 and 0.46 mm, with payload estimates of 0.779 and 0.836 kg. Under 1 kHz band-limited excitation, tuned SG differentiation is therefore the derivative estimator of SAPC; a learned component inside a twin needs a confidence measure of the kind discussed in [28] before it is trusted. A black-box alternative fared no better: an MLP with two hidden layers of 64 units on the forward-dynamics residual of the nominal twin fitted the training trajectories but not the held-out ones (validation NRMSE 0.60-0.72), and its twin replayed at 13.9-16.7 mm, no better than the nominal twin.

B. Full identification benchmark

Table 2 keeps the schemes needed for the argument. The complete benchmark runs thirteen schemes at the three sensor-noise levels with SG derivatives and once with simulator-exact derivatives, on the same three training and two held-out trajectories; Table 9 lists them at noise $\times 1$. Every calibrated twin is instantiated natively after the pseudo-inertia projection [7], except the LMI-constrained dense fit, whose bodies are consistent by construction.

Three observations complete Section 5.2. First, every scheme that uses the full library reaches the same linear-library NRMSE, whereas the restricted fits cannot represent the link-3 and friction terms; the payload-only fit attributes the friction discrepancy to an 0.86 kg payload with a centre of mass 37 mm off. Second, behavioral fidelity separates schemes through numerical stability rather than physics: projecting physically invalid link inertias after a dense bounded fit changes the torque function the fit explained, and group-norm thresholding cannot shrink aliased groups, so both replay at a fraction of a millimetre or worse. Third, physical attribution collapses in every scheme but the group selections: dense fits report 0.13 kg at every noise level and with any of the three fitting routines, and the elementwise sparse fits between 0.24 and 2.45 kg, whereas SAPC and the exhaustive search return 0.78-0.80 kg.

Table 10 pairs the dense variants with SAPC on the same three noise draws per condition, within one simulator and against the Isaac Sim reference. The LMI-constrained fit is physically consistent without projection, every pseudo-inertia eigenvalue of its bodies satisfying the 10^{-6} floor of the constraint, and all four schemes reach the same executable-twin NRMSE (0.024-0.026 within MuJoCo, 0.098-0.103 against Isaac Sim) and similar replay errors;

Table 9

Full same-simulator benchmark at noise $\times 1$: linear-library and executable-twin torque NRMSE, replay TCP RMSE and 0.2s open-loop divergence, payload mass (true 0.8 kg), payload centre-of-mass error and number of groups with a non-zero correction.

Scheme	NRMSE lib.	NRMSE twin	TCP [mm]	Open loop [mm]	Payload [kg]	CoM err. [mm]	Groups
Nominal twin	1.000	1.000	14.43	36.62	0.000	-	0
LS, payload group only	0.326	0.329	2.65	11.24	0.861	37	1
LS, payload and joint Coulomb/viscous	0.064	0.054	0.48	1.13	0.830	19	8
Dense LS (ridge 10^{-6})	0.041	0.025	0.027	0.077	0.130	368	15
Dense, SAPC fit	0.041	0.025	0.028	0.070	0.130	369	15
Dense, pseudo-inertia LMI	0.041	0.024	0.030	0.073	0.130	369	15
Dense bounded LS, projected	0.042	0.049	0.59	1.39	-0.314	390	15
Elementwise STLSQ	0.042	0.024	0.029	0.077	0.241	124	9
Bounded elementwise sparse	0.042	0.025	0.035	0.088	1.112	67	8
Group-norm STLSQ	0.042	0.044	0.53	1.21	-0.314	390	8
SAPC	0.041	0.024	0.030	0.056	0.794	1.3	4
Exhaustive best-subset	0.041	0.024	0.030	0.056	0.794	1.3	4
MLP residual	0.605	-	14.43	14.40	-	-	-
True parameters	0.042	0.024	0.027	0.042	0.800	0.0	4

Table 10

Paired comparison of dense variants and SAPC on the same data: payload mass (true 0.8 kg) and replay TCP RMSE, each the mean over three noise draws, for the MuJoCo (MJ) and Isaac Sim (IS) references at noise $\times 1$ and $\times 3$. The true-parameter twin does not depend on the noise draw.

Scheme	Payload [kg]				Replay TCP RMSE [mm]			
	MJ $\times 1$	MJ $\times 3$	IS $\times 1$	IS $\times 3$	MJ $\times 1$	MJ $\times 3$	IS $\times 1$	IS $\times 3$
SAPC	0.795	0.794	0.795	0.790	0.029	0.041	0.26	0.27
Dense LS	0.131	0.130	0.130	0.130	0.026	0.065	0.26	0.27
Dense, SAPC fit	0.130	0.130	0.130	0.130	0.026	0.053	0.26	0.27
Dense, pseudo-inertia LMI	0.130	0.131	0.130	0.130	0.026	0.051	0.26	0.27
True parameters	0.800	0.800	0.800	0.800	0.027	0.027	0.26	0.26

the payload estimate of every dense variant is 0.13 kg in every condition, with a standard deviation over draws of at most 0.006 kg for any scheme. Physical consistency and correct attribution are therefore different properties: the true correction and the dense one are both realisable bodies, and only the sparse-change prior chooses between them. With the Isaac Sim reference the executable-twin NRMSE exceeds the linear-library value, because the reference applies its friction in the library's analytic form while the twin keeps MuJoCo's constraint-based friction loss, whose knee differs at low speed; on the rows above 0.1 rad/s the executable-twin NRMSE drops to 0.028.

C. Cross-simulator implementation details

Software and model Every reference and twin is generated from the Menagerie FR3 description [33] and computed with MuJoCo 3.13.0; the Gazebo reference runs in Gazebo Harmonic with its DART physics plugin [35] and the Isaac Sim reference in Isaac Sim 4.5 [34]. Exact software versions, the conversion scripts and the simulator bridges are in the repository.

Gazebo reference The Gazebo worlds are SDF files generated from the MJCF, with the inertials (mass, centre of mass, principal inertia and its orientation), joint viscous damping and Coulomb friction copied verbatim and the link poses baked at the home configuration. A nominal world and one with the discrepancies of Table 1 are generated, each with and without the plate and spherical probe, and the free-space worlds carry no collision elements. A 2 s hold at home reproduced the MuJoCo gravity torques of joints 2 and 4 ($-26.02/19.06$ against $-26.00/19.03$ N m) with a tracking error of 8×10^{-5} rad, which validates the conversion.

Missing rotor inertia and gain adjustment SDF has no rotor-inertia element, and deployed Franka Gazebo models carry none. Folding the armature into the link inertia about the joint axis, the usual workaround, is rejected by sdfORMAT because it violates the triangle inequality on the slender wrist links ($0.074 \gg 0.0002 + 0.0002 \text{ kg m}^2$): no physical body has that inertia. Without armature the effective inertia of joint 7 is about 10^{-4} kg m^2 , 750 times smaller than in MuJoCo, and the 1 kHz discrete PD with $K_{d,7} = 4 \text{ N m s}$ is unstable ($K_d \Delta t / I \approx 33$): the torque saturated at $\pm 12 \text{ N m}$ in 77 % of the samples and reversed sign every millisecond. $K_{d,7}$ was therefore lowered to 0.05 N m s on every plant of the cross-simulator stages, reference and twins alike, and the missing rotor inertia was left for the identification to discover as the armature group.

Isaac Sim reference The Isaac Sim stage is generated from the same description as a native fixed-base articulation, and the 0.8 kg payload of the reference is merged into link 7 by the parallel-axis theorem, which is equivalent to a rigidly attached body. The nominal rotor inertia is kept as the articulation's armature, so this reference does not repeat the description difference of the SDF one. The reference friction is applied as an explicit passive torque $f_c \tanh(\dot{q}/0.02) + f_v \dot{q}$ at every physics step, because the joint-friction parameter of the Isaac Sim articulation interface used here is a dimensionless, load-dependent coefficient that does not correspond to a torque. This friction law is the library's own Coulomb and viscous form, whereas the MuJoCo twin keeps its constraint-based friction loss, so the Isaac Sim comparison tests a different articulation solver and contact model rather than a foreign friction law. Before collection the kinematics, mass matrix, gravity torques and one-step accelerations were checked against MuJoCo at two poses, and after collection the TCP of every sample was checked against MuJoCo forward kinematics to $20 \mu\text{m}$. The press uses the rigid contact of Isaac Sim with zero restitution, and no contact parameter is fitted.

Press geometry During the 1 s settle under the true load the probe sags by 13.5 mm in Gazebo and 15.5-16.0 mm in MuJoCo, which sets the 26 mm plate gap and 29 mm descent of every press in the paper.

D. Contact metric sensitivity

The press metrics of Table 6 are the onset (first sample with normal force above 0.05 N), the holding force (mean over 2.2-2.9 s, inside the 1 s hold), the impulse (integral of the normal force over the press) and the TCP RMSE. Peak force and full-waveform RMSE are reported only here, because both are dominated by the contact solver rather than by the robot parameters. The reference plate is stiff (solref time constant 2 ms) and its force waveform chatters at about 100 Hz and above, with a standard deviation of about 3 N during the hold. Because the chatter phase is not reproducible, the waveform RMSE stays at 5.1 N even for the true parameters with the true plate and the matched 0.5 ms implicit integrator, whose TCP RMSE is 0.018 mm; it is 5.7 N for the true parameters with the twins' 1 ms implicit-fast integrator, 5.1-5.3 N for the three calibrated twins and 7.1 N for the nominal twin. The impact peak is more integrator-sensitive still: with the true parameters it is 126 N under the matched integrator (reference 126 N) but 162 N at 1 ms implicit-fast, whereas onset, holding force and impulse move by 0 ms, 0.15 N and 0.14 N s between the two integrators. Onset, holding force, impulse and TCP error are therefore the primary contact metrics; a validation scoring pointwise force error alone, as output-level trajectory validation of co-simulation twins does [3], would place every twin of Table 6, the true parameters included, in the same 5-7 N band, and choosing metrics the solver cannot dominate is part of the validation discipline formalised for robot simulations in [9].

Contact parameters are not identified by this press. A grid search over plate solref time constants of 1, 1.5, 2, 3, 5 and 8 ms and damping ratios of 0.3, 0.6, 1.0 and 1.5, minimising the force RMSE, selected 8 ms and 1.5 for every calibrated arm and for the nominal arm, against the true 2 ms and 0.6. For the SAPC arm it lowered the force RMSE from 5.2 to 4.6 N by smoothing the chatter, moved the peak away from the reference (114 to 88 N), left onset, holding force and impulse nearly unchanged (1.486 s, 10.77 N, 18.36 N s) and raised the TCP RMSE from 0.73 to 0.81 mm. The maximum penetration in the reference is 0.044 mm, so the press excites the plate compliance far too little to identify it; this is the limitation stated in Section 6.5, and every result in the paper uses the nominal contact parameters.

References

- [1] G. L. Stavrinos, H. D. Karatza, Cyber-physical systems, digital twins and Industry 4.0: The role of modeling and simulation, *Simulation Modelling Practice and Theory* 124 (2023) 102727. doi:10.1016/j.simpat.2023.102727.
- [2] G. Wang, Z. Zhang, A. Nazarova, A digital twin-based platform for testing and optimization of path planning algorithms, *Simulation Modelling Practice and Theory* 145 (2025) 103206. doi:10.1016/j.simpat.2025.103206.

- [3] I. Ahmad, B. Siddique, M. A. Islam, Z. U. Haq, Z. Niu, M. M. Waqas, Q. Yang, Z. Qiu, Development and evaluation of digital twin model for rack and pinion drive vegetable seedling transmission device using Adams/MATLAB co-simulation, *Simulation Modelling Practice and Theory* 142 (2025) 103132. doi:10.1016/j.simpat.2025.103132.
- [4] A. Allevato, E. S. Short, M. Pryor, A. Thomaz, TuneNet: One-shot residual tuning for system identification and sim-to-real robot task transfer, in: L. P. Kaelbling, D. Kragic, K. Sugiura (Eds.), *Proceedings of the Conference on Robot Learning (CoRL 2019)*, Vol. 100 of *Proceedings of Machine Learning Research*, PMLR, 2020, pp. 445–455, arXiv:1907.11200.
- [5] D. Bargellini, A. Pasquali, A. Govoni, R. Zanella, G. Palli, Enhancing Sim2Real transfer for torque-controlled robots through Real2Sim dynamics estimation and reinforcement learning, arXiv:2608.22629, presented at the 2026 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM) (2026).
- [6] C. Gaz, M. Cognetti, A. Oliva, P. Robuffo Giordano, A. De Luca, Dynamic identification of the Franka Emika Panda robot with retrieval of feasible parameters using penalty-based optimization, *IEEE Robotics and Automation Letters* 4 (4) (2019) 4147–4154. doi:10.1109/LRA.2019.2931248.
- [7] P. M. Wensing, S. Kim, J.-J. E. Slotine, Linear matrix inequalities for physically consistent inertial parameter identification: A statistical perspective on the mass distribution, *IEEE Robotics and Automation Letters* 3 (1) (2018) 60–67. doi:10.1109/LRA.2017.2729659.
- [8] ASME, Assessing credibility of computational modeling through verification and validation: Application to medical devices, Tech. Rep. V&V 40-2018, American Society of Mechanical Engineers (2018).
- [9] P. J. Durst, D. McInnis, J. Davis, C. T. Goodin, A novel framework for verification and validation of simulations of autonomous robots, *Simulation Modelling Practice and Theory* 117 (2022) 102515. doi:10.1016/j.simpat.2022.102515.
- [10] S. Xie, W. Zhang, M. Shen, F. Xue, A modular configuration-based rapid verification and validation framework combining analytical and simulation methods for intelligent manufacturing systems, *Simulation Modelling Practice and Theory* 142 (2025) 103136. doi:10.1016/j.simpat.2025.103136.
- [11] J. Shi, Q. Zhou, C. Pan, A credibility evaluation method for digital twin based on improved evidence theory, *Simulation Modelling Practice and Theory* 143 (2025) 103152. doi:10.1016/j.simpat.2025.103152.
- [12] A. Farley, J. Wang, J. A. Marshall, How to pick a mobile robot simulator: A quantitative comparison of CoppeliaSim, Gazebo, MORSE and Webots with a focus on accuracy of motion, *Simulation Modelling Practice and Theory* 120 (2022) 102629. doi:10.1016/j.simpat.2022.102629.
- [13] G. D. Wijaya, W. Caesarendra, M. I. Petra, G. Królczyk, A. Glowacz, Comparative study of Gazebo and Unity 3D in performing a virtual pick and place of Universal Robot UR3 for assembly process in manufacturing, *Simulation Modelling Practice and Theory* 132 (2024) 102895. doi:10.1016/j.simpat.2024.102895.
- [14] M. Gautier, W. Khalil, Direct calculation of minimum set of inertial parameters of serial robots, *IEEE Transactions on Robotics and Automation* 6 (3) (1990) 368–373. doi:10.1109/70.56655.
- [15] W. Khalil, E. Dombre, *Modeling, Identification and Control of Robots*, Butterworth-Heinemann, Oxford, UK, 2004, first published by Hermes Penton Science, London, 2002.
- [16] P. M. Wensing, G. Niemeyer, J.-J. E. Slotine, A geometric characterization of observability in inertial parameter identification, *The International Journal of Robotics Research* 43 (14) (2024) 2274–2302. doi:10.1177/02783649241258215.
- [17] M. C. Kennedy, A. O'Hagan, Bayesian calibration of computer models, *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 63 (3) (2001) 425–464. doi:10.1111/1467-9868.00294.
- [18] J. Brynjarsdóttir, A. O'Hagan, Learning about physical parameters: The importance of model discrepancy, *Inverse Problems* 30 (11) (2014) 114007. doi:10.1088/0266-5611/30/11/114007.
- [19] C. D. Sousa, R. Cortesão, Physical feasibility of robot base inertial parameter identification: A linear matrix inequality approach, *The International Journal of Robotics Research* 33 (6) (2014) 931–944. doi:10.1177/0278364913514870.
- [20] C. G. Atkeson, C. H. An, J. M. Hollerbach, Estimation of inertial parameters of manipulator loads and links, *The International Journal of Robotics Research* 5 (3) (1986) 101–119. doi:10.1177/027836498600500306.
- [21] S. L. Brunton, J. L. Proctor, J. N. Kutz, Discovering governing equations from data by sparse identification of nonlinear dynamical systems, *Proceedings of the National Academy of Sciences* 113 (15) (2016) 3932–3937. doi:10.1073/pnas.1517384113.
- [22] Z. Chen, Y. Liu, H. Sun, Physics-informed learning of governing equations from scarce data, *Nature Communications* 12 (1) (2021) 6136. doi:10.1038/s41467-021-26434-1.
- [23] T. N. Truong, S. Kim, High-performance fixed-time active fault-tolerant control for robotic manipulators via sparse physics-informed dynamics learning, *Information Sciences* 758 (2027) 124001. doi:10.1016/j.ins.2026.124001.
- [24] C. E. Mower, R. Zong, H. Bou-Ammar, Data-driven interpretable hybrid robot dynamics, arXiv:2512.11900 (2025). arXiv:2512.11900.
- [25] R. D. Yadav, S. Ujjawal, S. Sun, S. Roy, W. Pan, Physics-aware sparse learning and selective online adaptation for Euler-Lagrange robot dynamics, arXiv:2606.09640 (2026). arXiv:2606.09640.
- [26] M. Heravi, A. Molaei, R. Ghabcheloo, Physics-informed data-driven modeling of rock motion dynamics in excavation using a high-fidelity simulator, *Simulation Modelling Practice and Theory* 145 (2025) 103208. doi:10.1016/j.simpat.2025.103208.
- [27] E. Heiden, D. Millard, E. Coumans, G. S. Sukhatme, Augmenting differentiable simulators with neural networks to close the sim2real gap, arXiv:2007.06045 (2020).
- [28] S. Chabanet, H. Bril El-Haouzi, P. Thomas, Active learning confidence measures for coupling strategies in digital twins integrating simulation and data-driven submodels, *Simulation Modelling Practice and Theory* 140 (2025) 103092. doi:10.1016/j.simpat.2025.103092.
- [29] R. G. Sargent, Verification and validation of simulation models, *Journal of Simulation* 7 (1) (2013) 12–24. doi:10.1057/jos.2012.20.
- [30] Y. Wang, Y. Huang, S. Viancha, M. B. Khamesee, Digital twin for magnetic levitation systems: General architecture design and uncertainty analysis, *Simulation Modelling Practice and Theory* 142 (2025) 103134. doi:10.1016/j.simpat.2025.103134.
- [31] T. Xia, J. Zuo, J. Ding, Y. Pan, Degradation-adaptive digital twin for long-term consistency, *Simulation Modelling Practice and Theory* 149 (2026) 103276. doi:10.1016/j.simpat.2026.103276.

- [32] E. Todorov, T. Erez, Y. Tassa, MuJoCo: A physics engine for model-based control, in: IEEE/RSJ International Conference on Intelligent Robots and Systems, 2012, pp. 5026–5033. doi:10.1109/IR0S.2012.6386109.
- [33] K. Zakka, Y. Tassa, MuJoCo Menagerie Contributors, MuJoCo Menagerie: A collection of high-quality simulation models for MuJoCo, http://github.com/google-deepmind/mujoco_menagerie (2022).
- [34] NVIDIA, Isaac Sim 4.5 documentation, <https://docs.isaacsim.omniverse.nvidia.com/4.5.0/> (2025).
- [35] J. Lee, M. X. Grey, S. Ha, T. Kunz, S. Jain, Y. Ye, S. S. Srinivasa, M. Stilman, C. K. Liu, DART: Dynamic animation and robotics toolkit, *Journal of Open Source Software* 3 (22) (2018) 500. doi:10.21105/joss.00500.